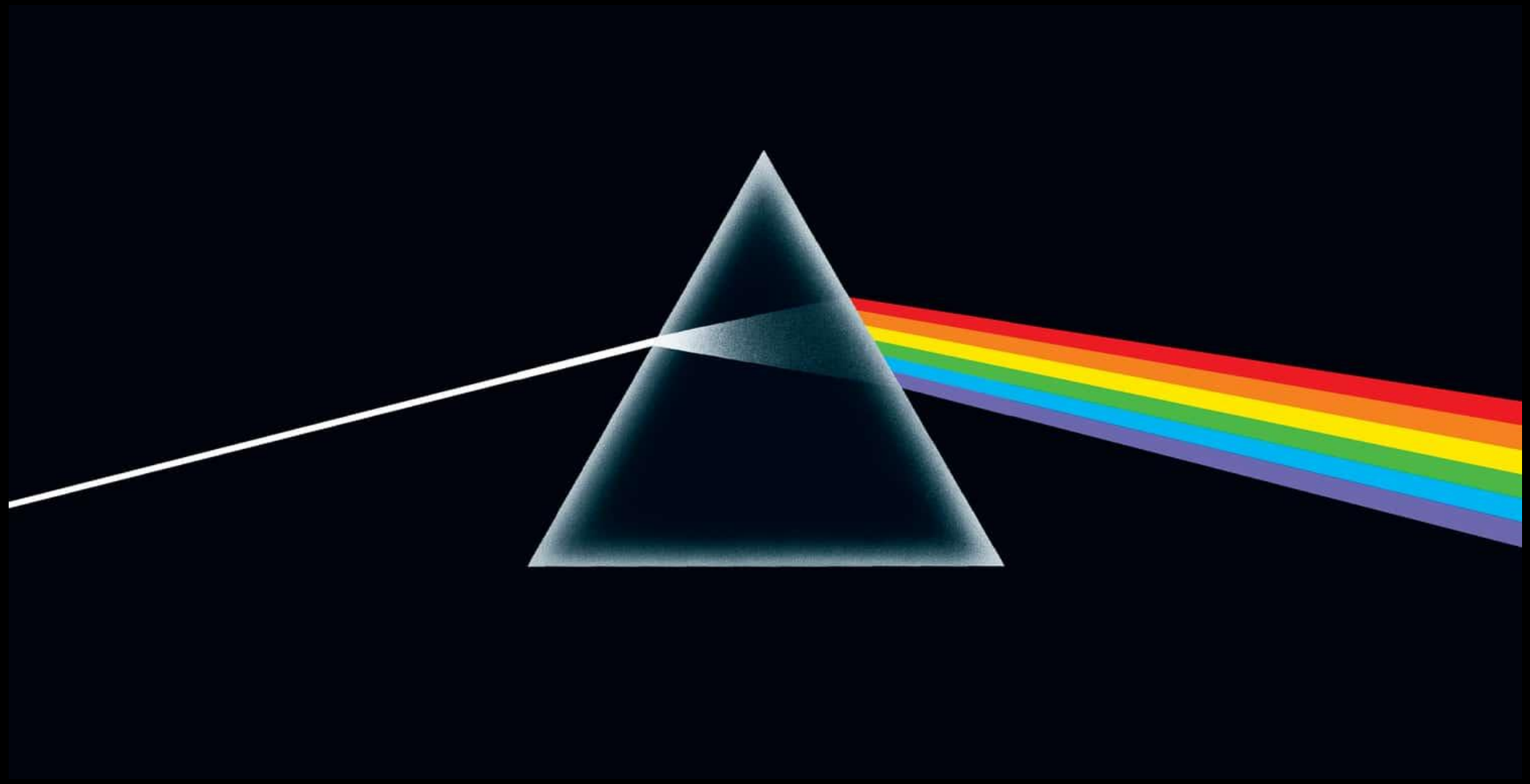
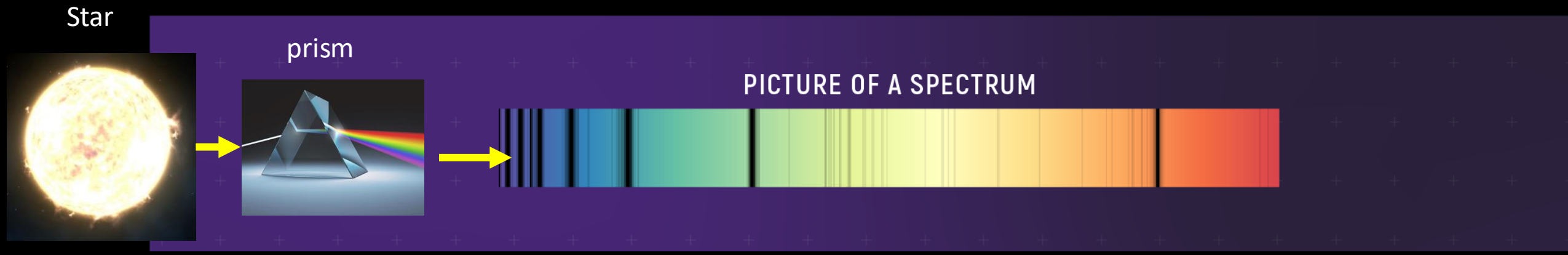
ASTR 302 Fall 2026: Spectroscopy



Spectroscopy



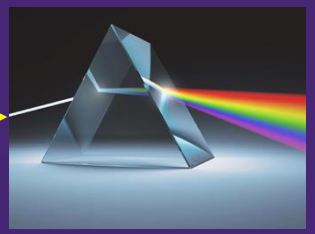
Spectroscopy is a technique used to study objects and their composition based the intensity of light as function of wavelength.

Figure from NASA

Star



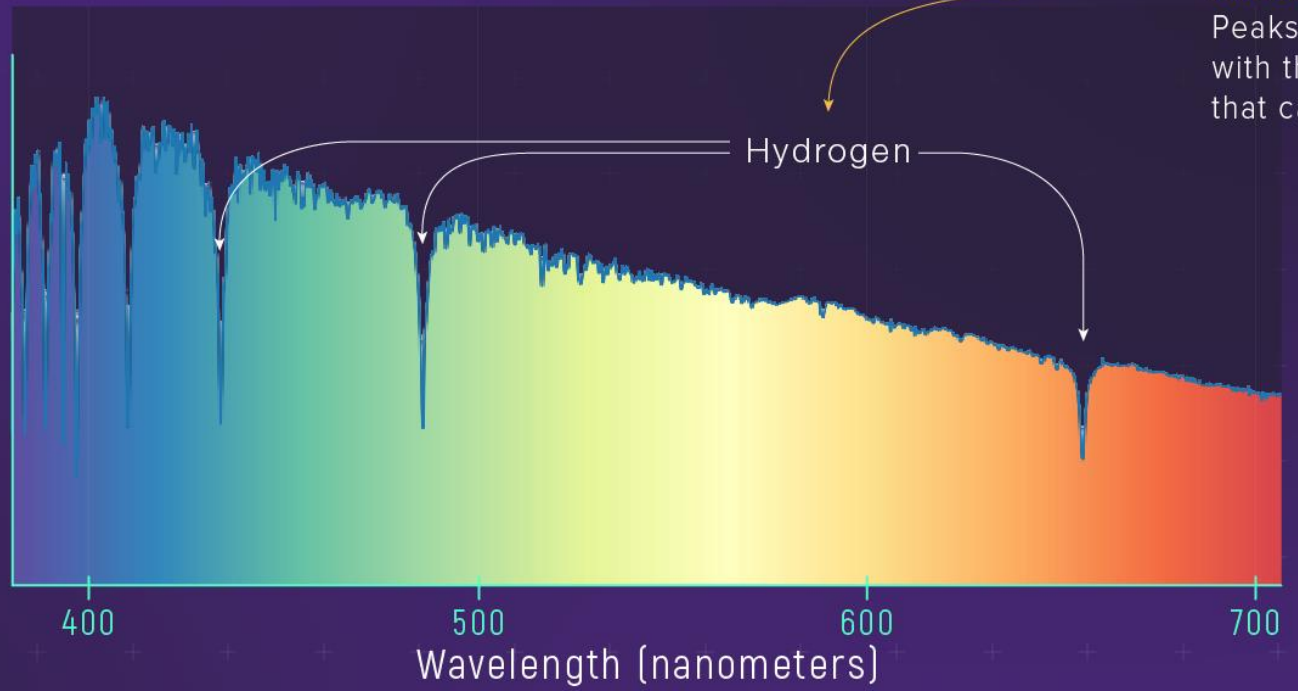
prism



PICTURE OF A SPECTRUM



GRAPH OF A SPECTRUM



Astronomer's interpretation:
Peaks and valleys are labeled with the elements and compounds that caused them.

Brightness
(might be labeled as intensity, counts, flux, power, absorbance, transmittance, or reflectance)

Color
(often labeled as wavelength, but can also be labeled as energy or frequency)

Figure from NASA

Spectroscopy: Composition

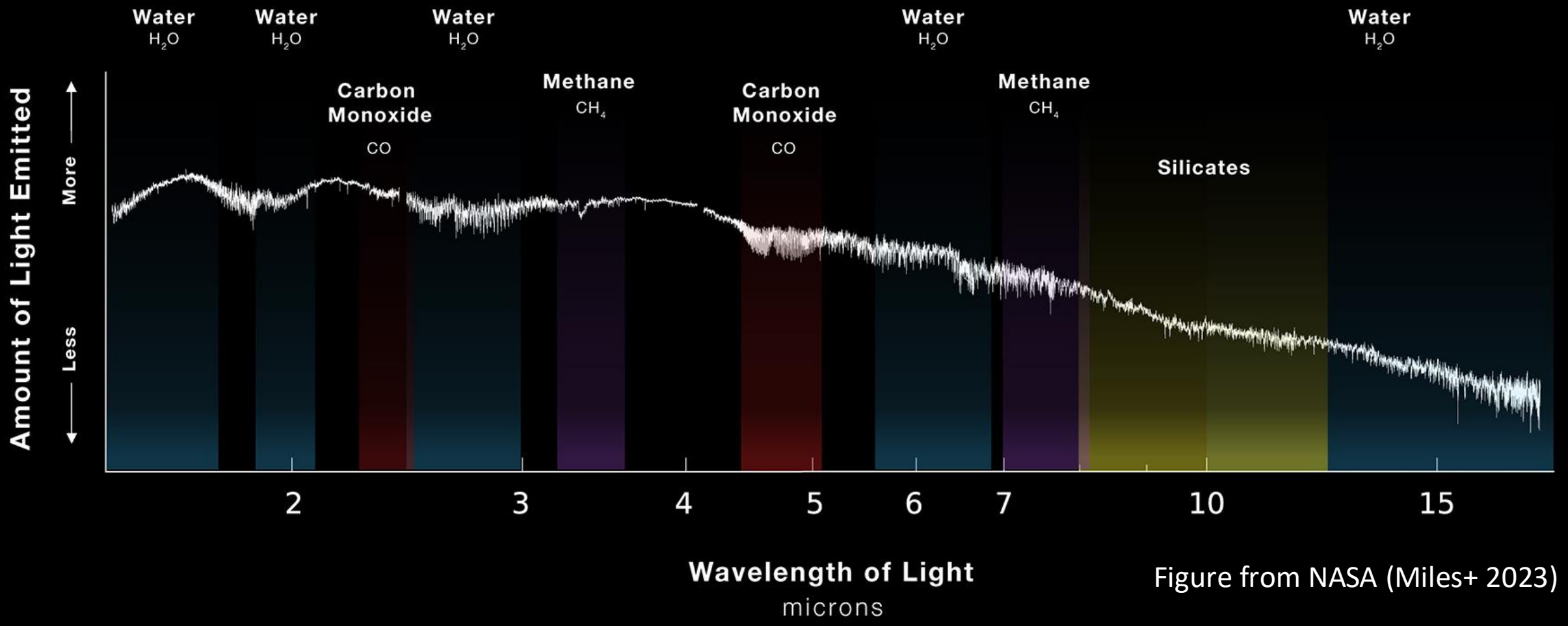


Figure from NASA (Miles+ 2023)

Light interacts with atoms and molecules (even isotopes) at distinct wavelengths.
We can understand the composition of various things in detail.

Spectroscopy: Composition

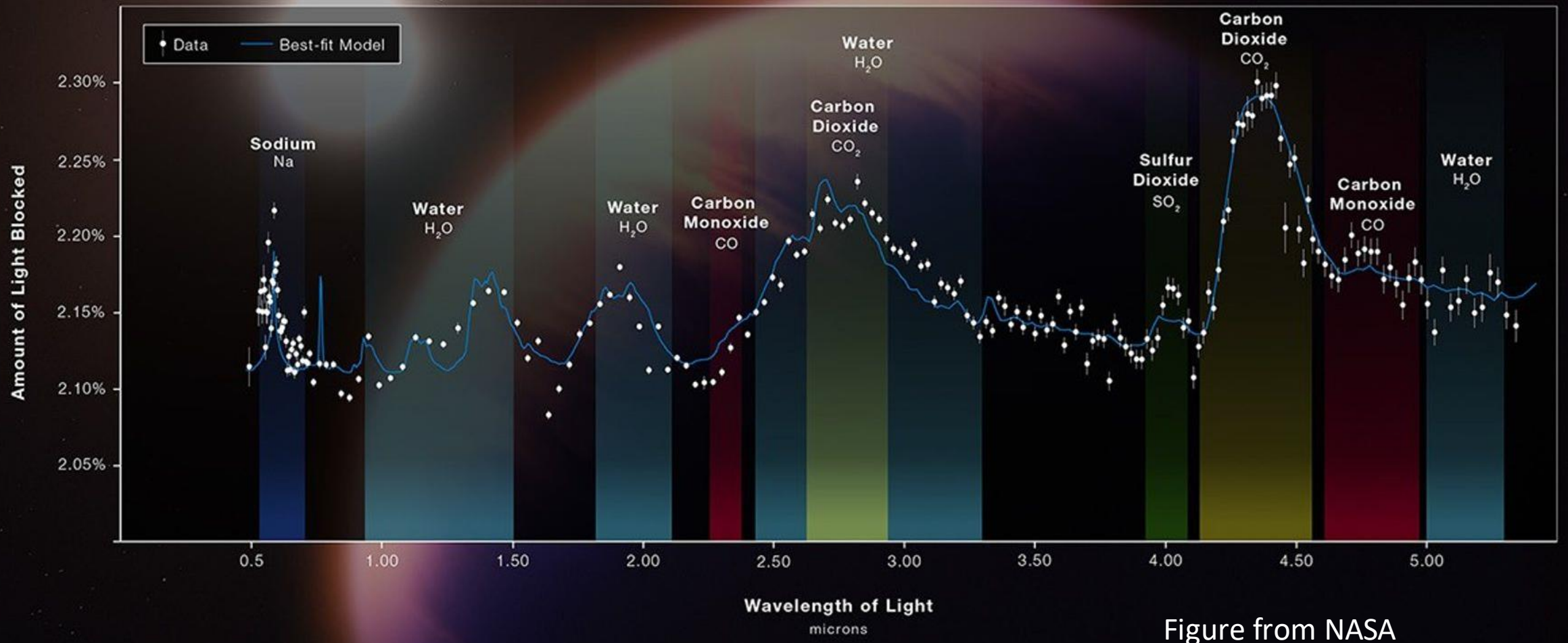
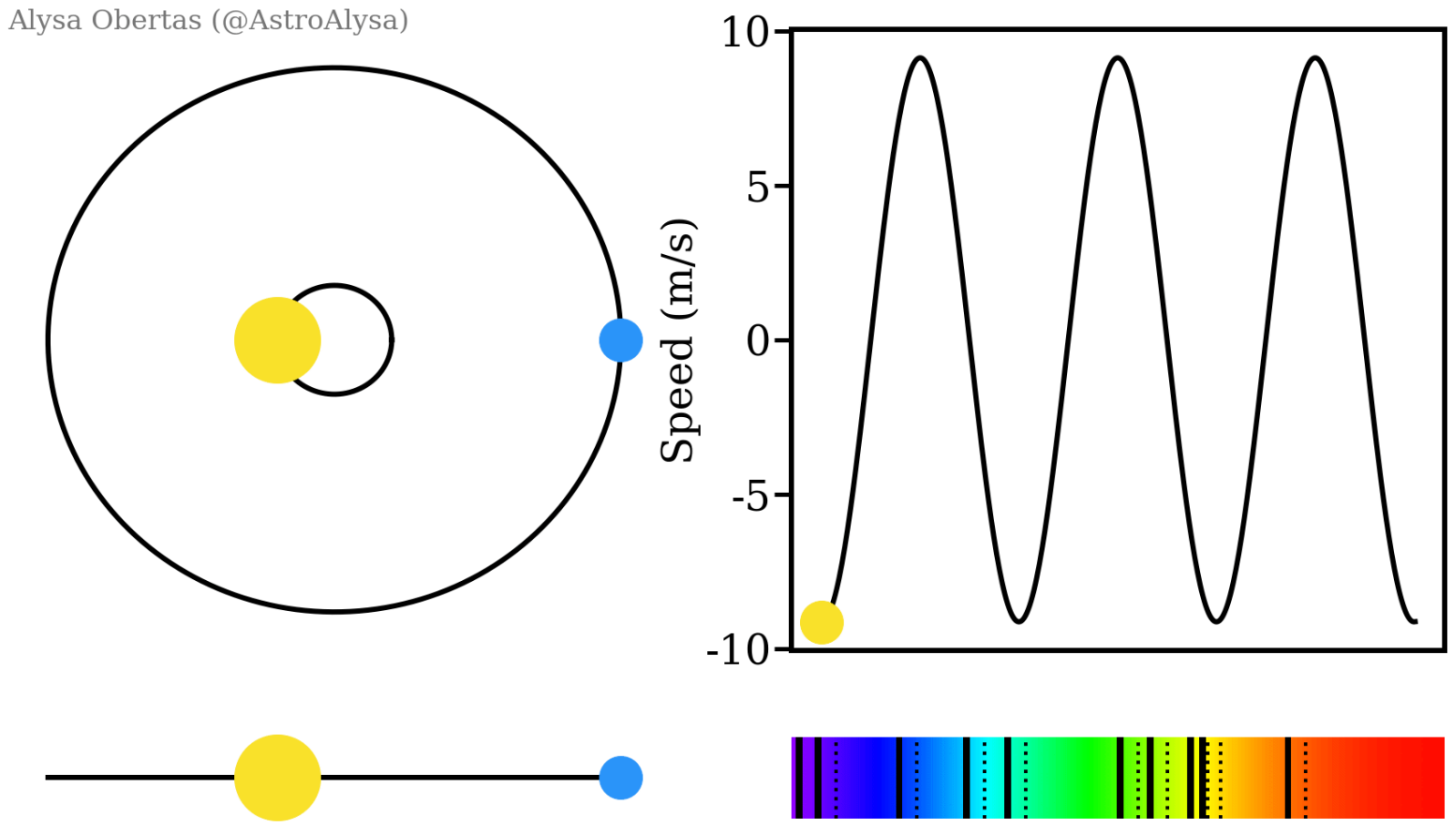


Figure from NASA

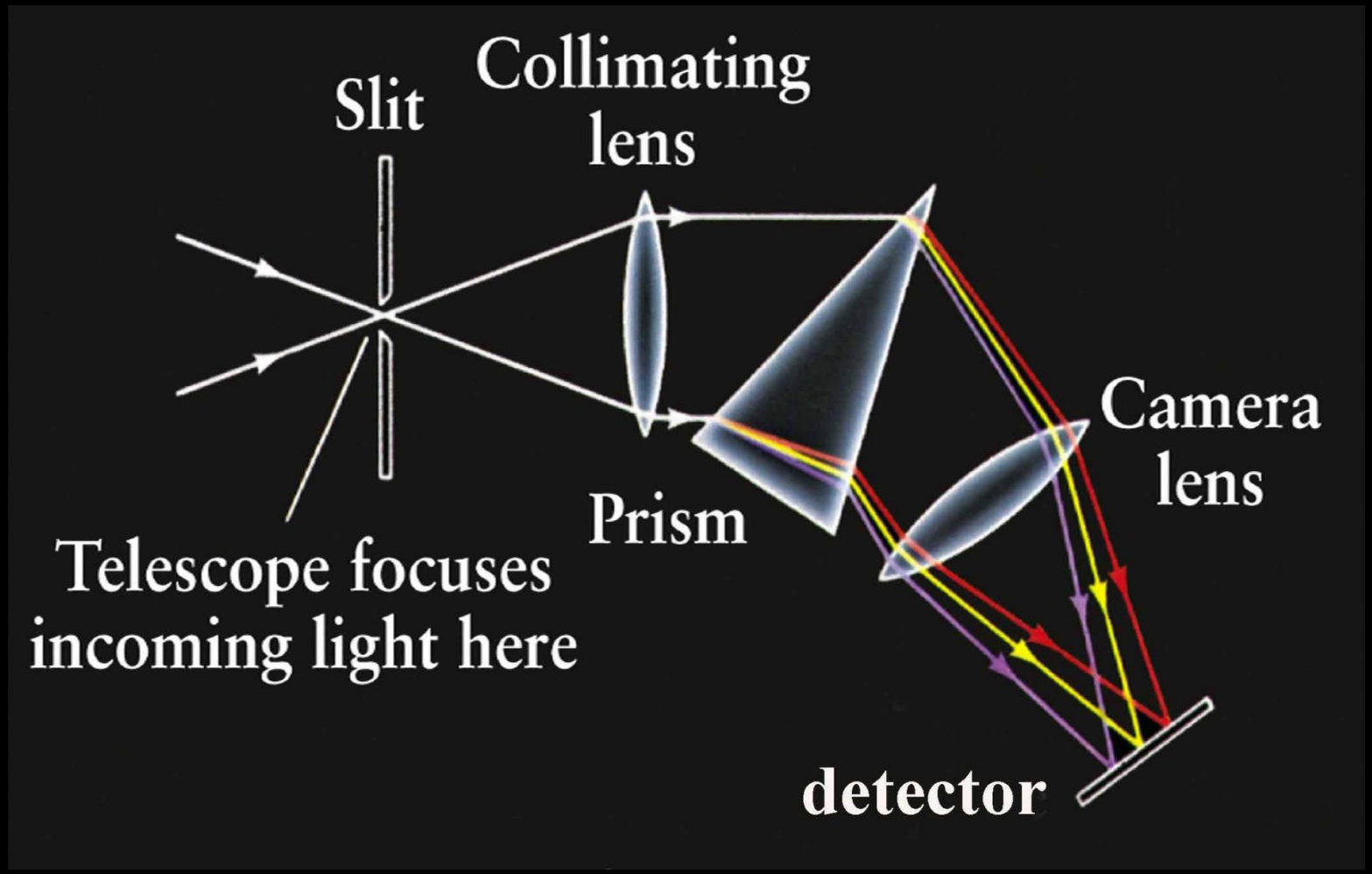
Light interacts with atoms and molecules (even isotopes) at distinct wavelengths. We can understand the composition of various things in detail.

Spectroscopy: Radial Velocities



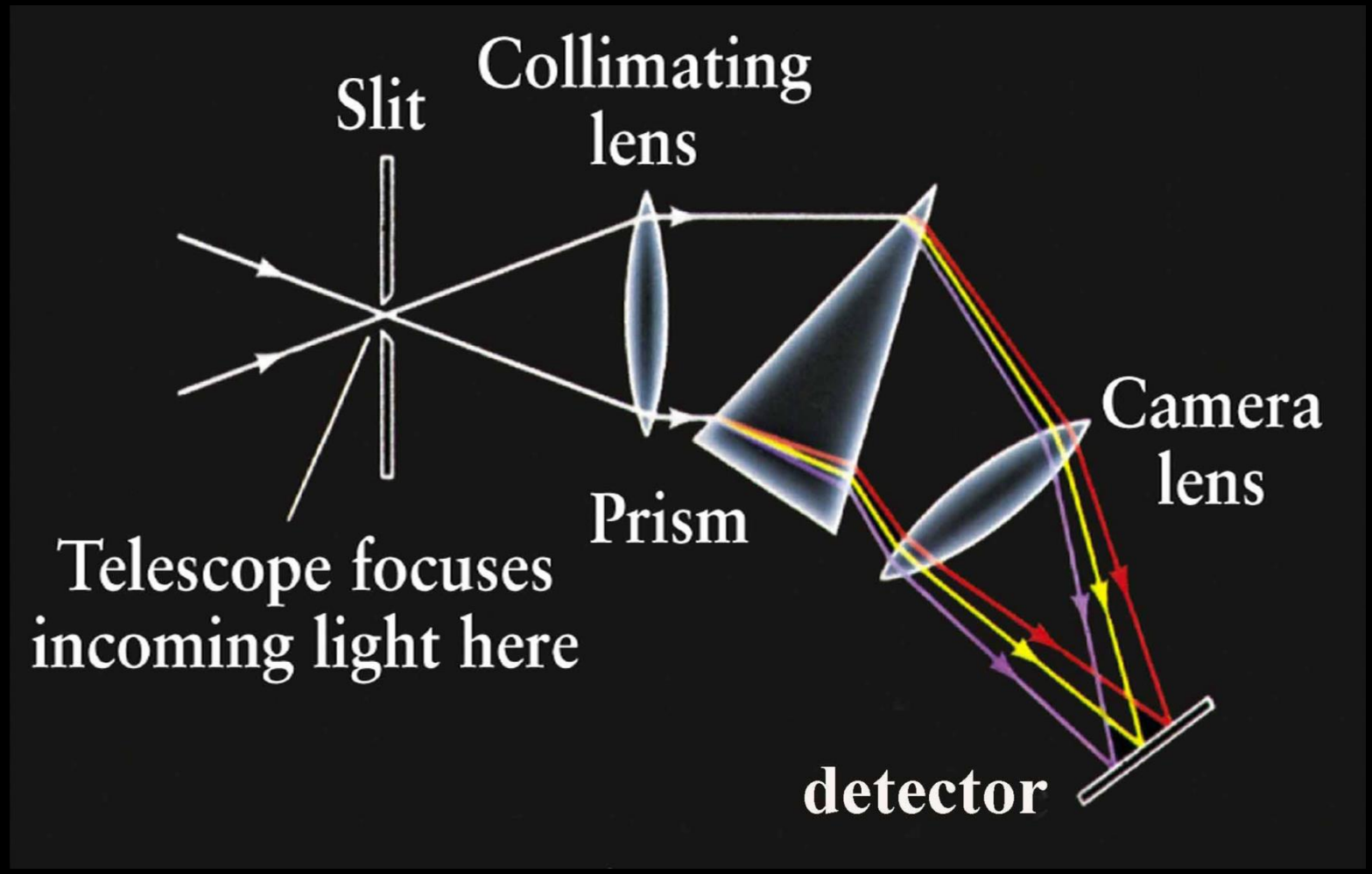
Spectroscopy can be used to measure velocity shifts in astronomical objects. This applies to exoplanets, circumstellar disks, galaxies, etc.

What is a Spectrograph



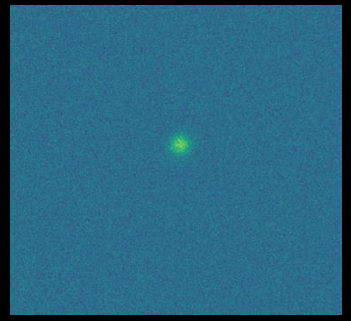
Spectrographs are instruments that split light into different wavelength components then records it.

Parts of a Spectrograph: Slit



- **Slit** – Limits the aperture of the spectrograph to focus on a specific target or area.

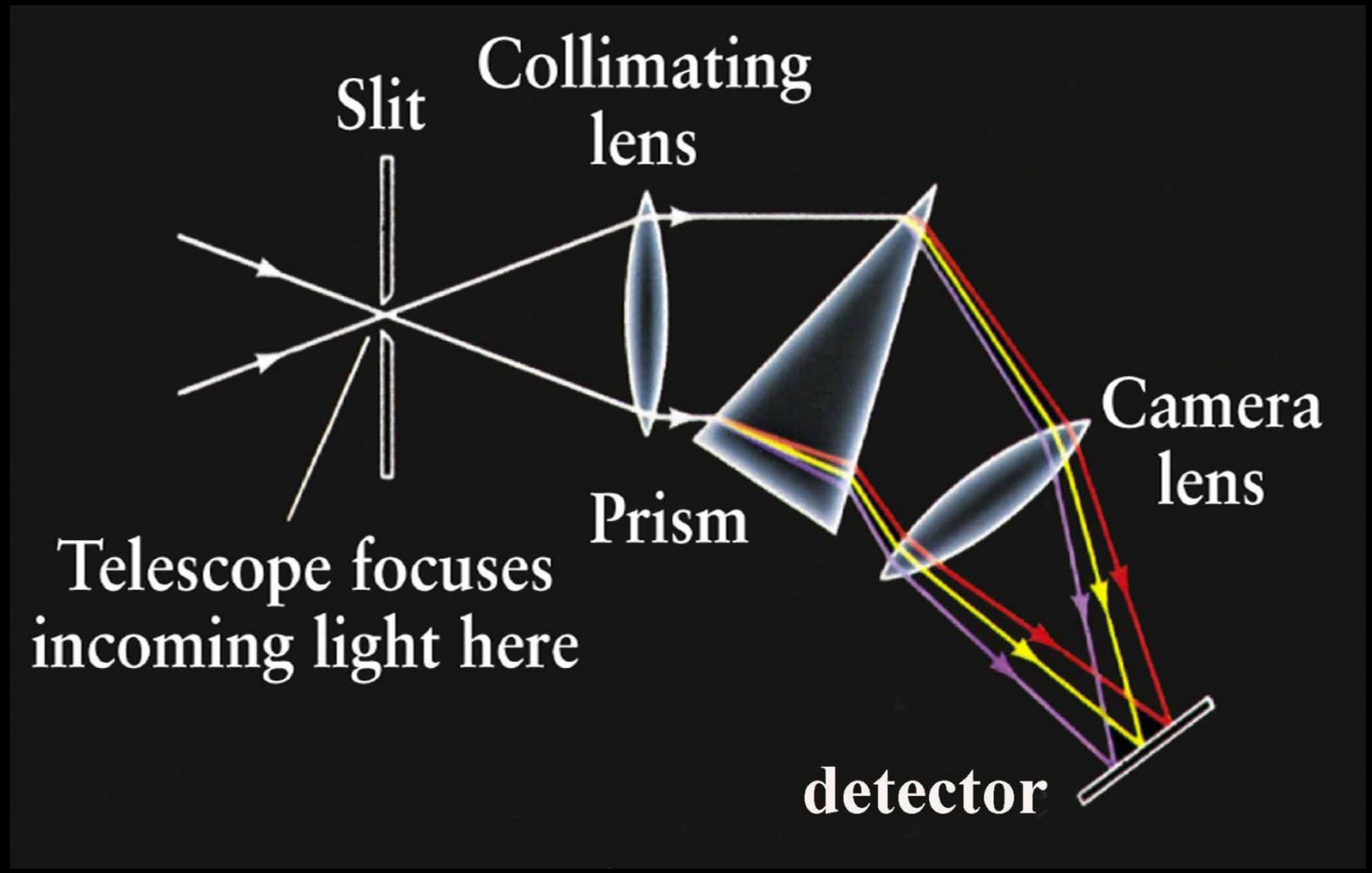
Focal Plane no Slit



Focal Plane w/ Slit

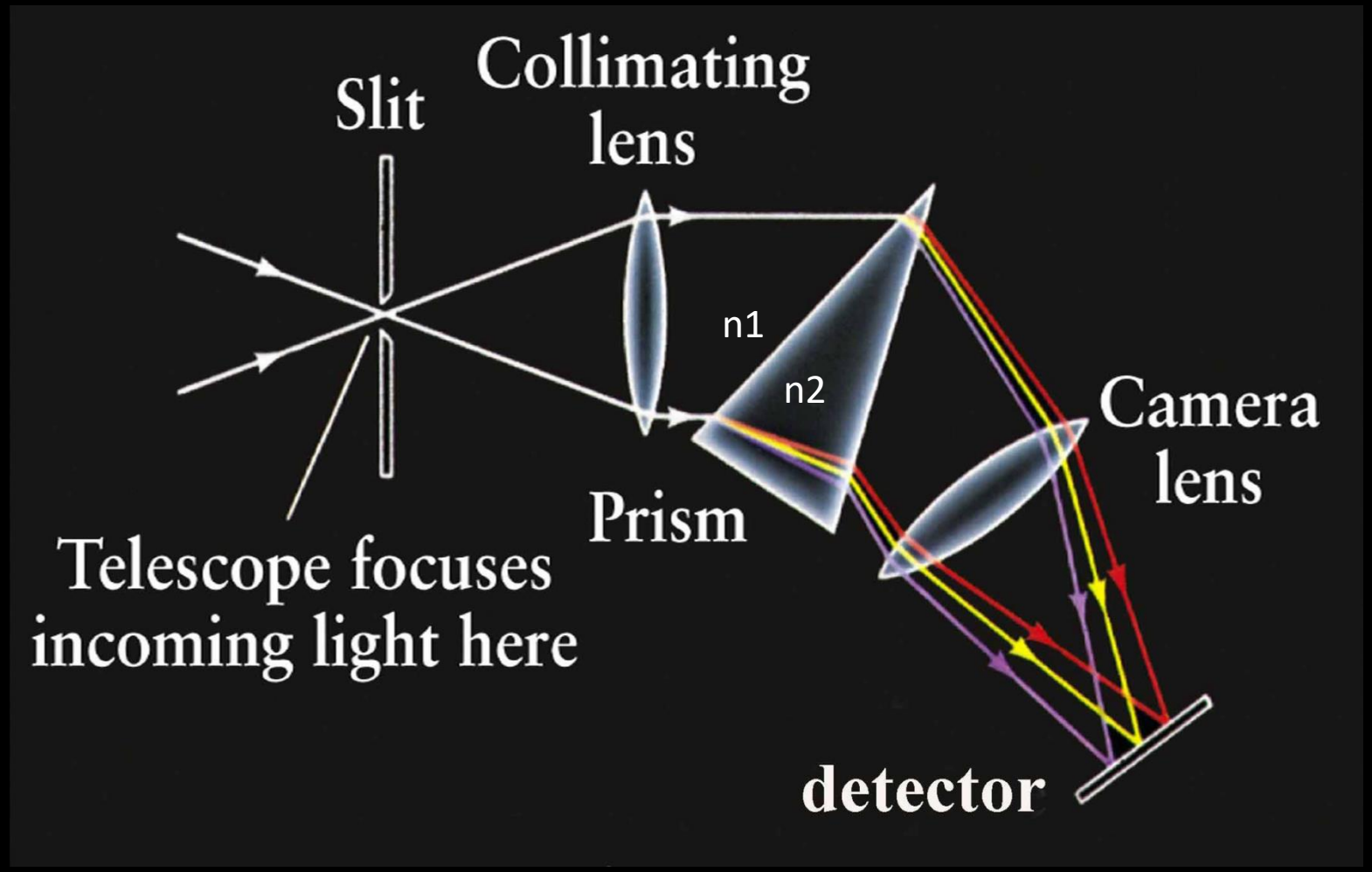


Parts of a Spectrograph: Collimator



- **Collimator** – Collimators diverts light onto a parallel path.
- Refraction is a wavelength dependent phenomena, therefore you want your light rays parallel before dispersion.

Parts of a Spectrograph: Dispersive Element

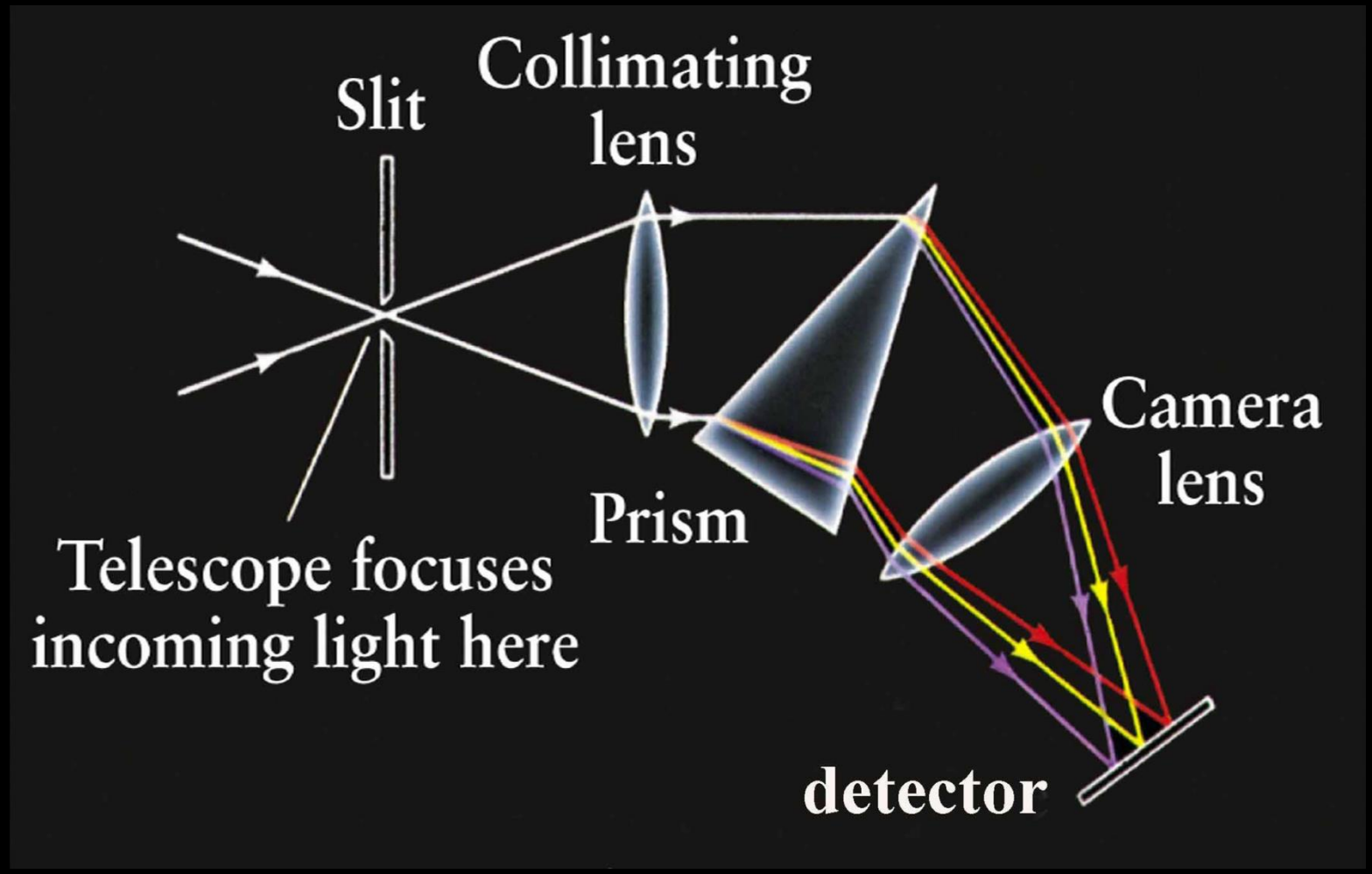


- **Dispersive Element** - All spectrographs have an optic that can separate light by wavelength. The simplest spectrographs use **prisms**.
- Prisms make use of the fact that index of refraction is wavelength dependent.

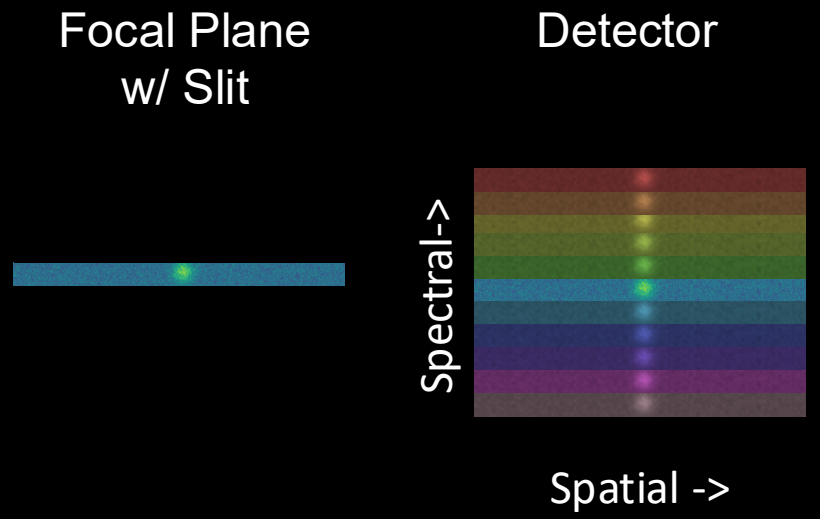
Schott Dispersion Equation:

$$n^2 = A_0 + A_1\lambda^2 + \frac{A_2}{\lambda^2} + \frac{A_3}{\lambda^4} + \frac{A_4}{\lambda^6} + \frac{A_5}{\lambda^8}$$

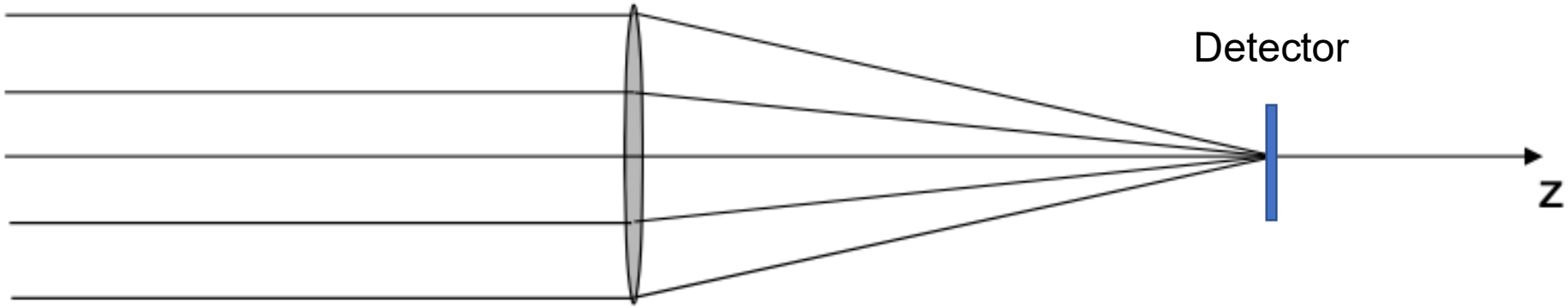
Parts of a Spectrograph: Imaging Lens + Camera



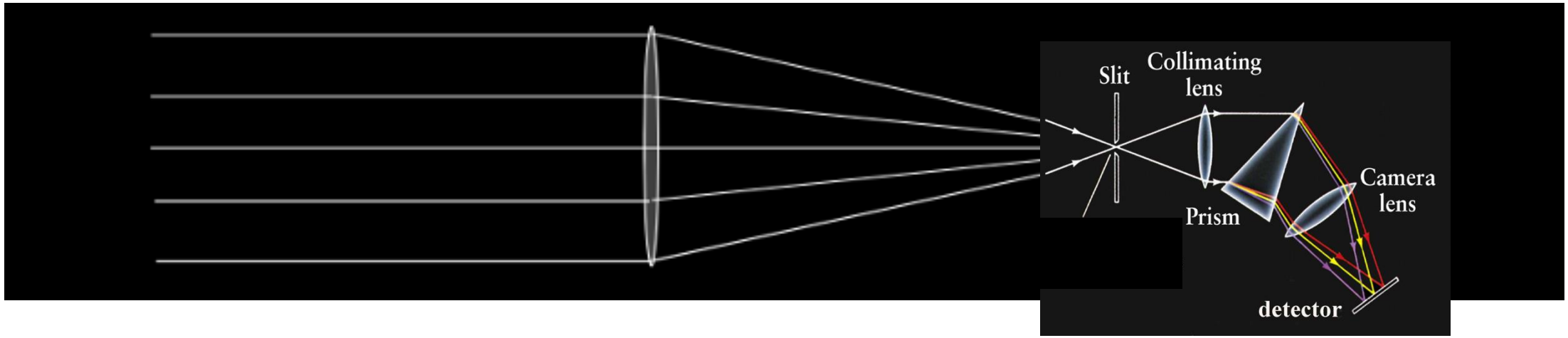
- Another lens is used with a detector to image the dispersed light leaving the prism.



Ray Diagrams Including Telescope



Ray diagram of an Imaging Camera



Ray diagram of a Spectrograph

Boller & Chivens Spectrograph

https://bollerandchivens.com/

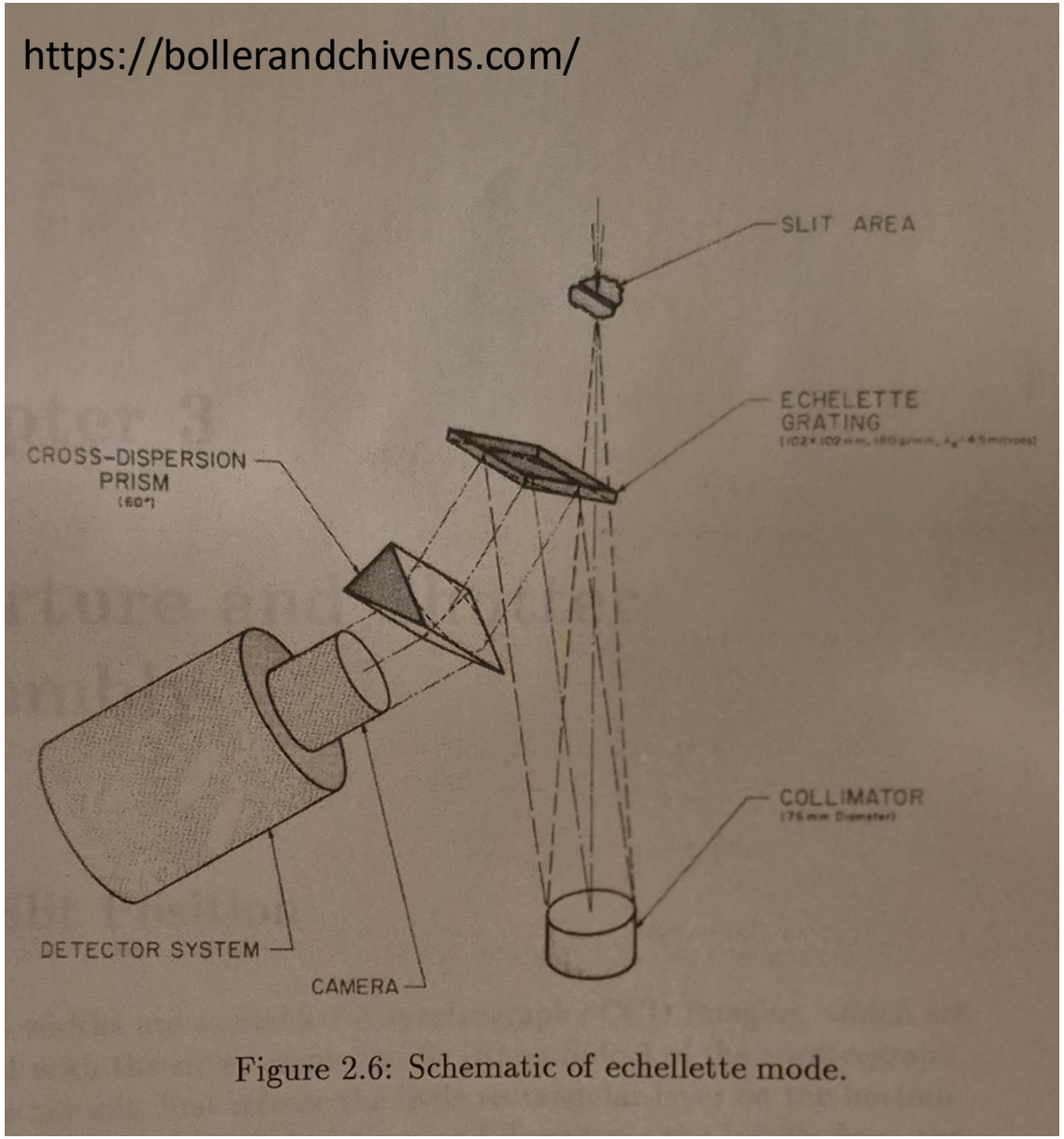


Figure 2.6: Schematic of echellette mode.

Boller & Chivens Spectrograph

Instrument

https://bollerandchivens.com/

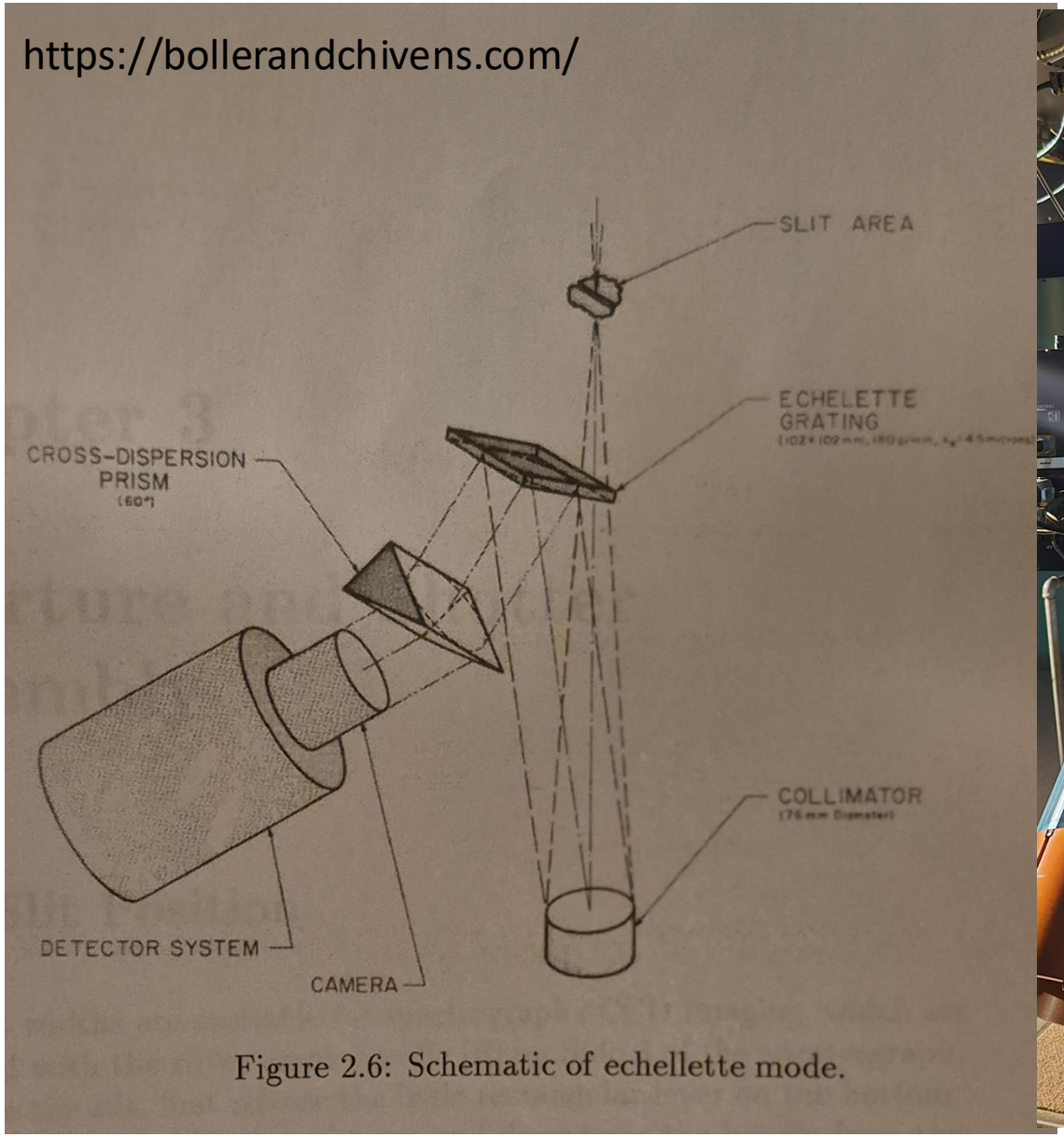
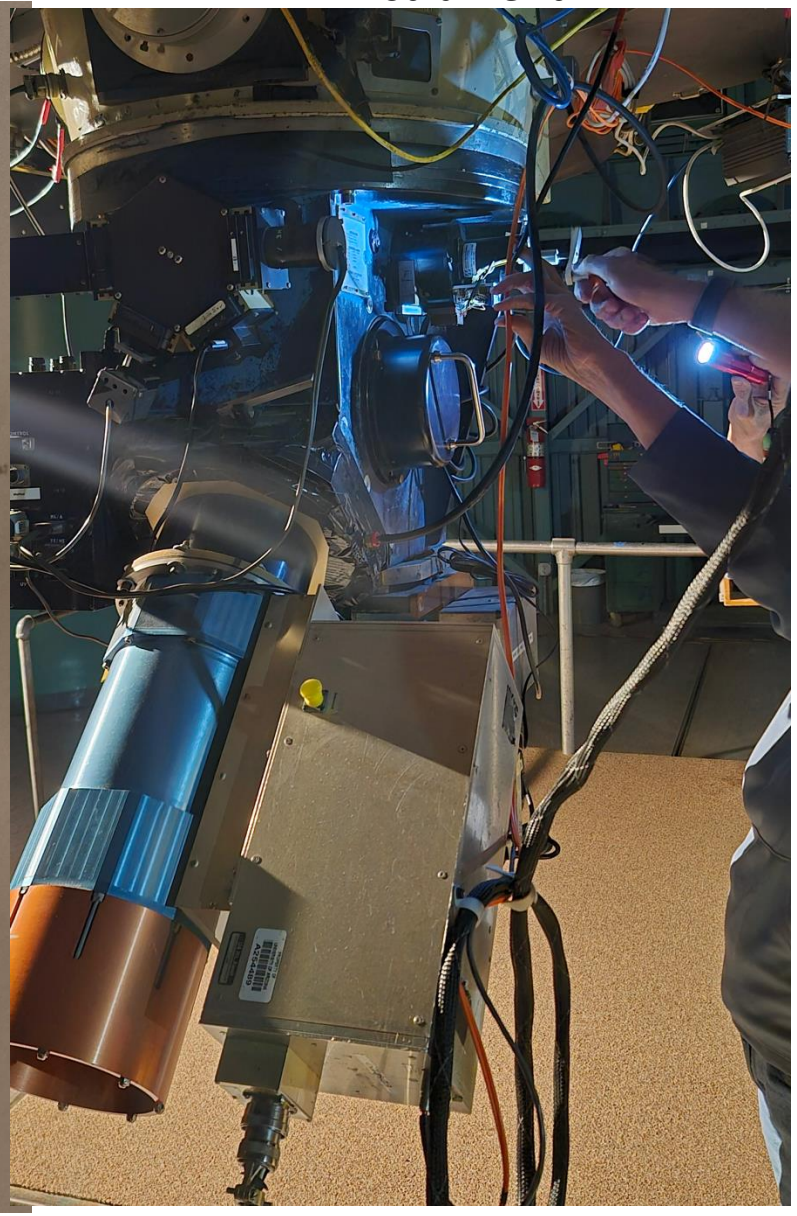


Figure 2.6: Schematic of echellette mode.





Boller & Chivens Spectrograph

https://bollerandchivens.com/

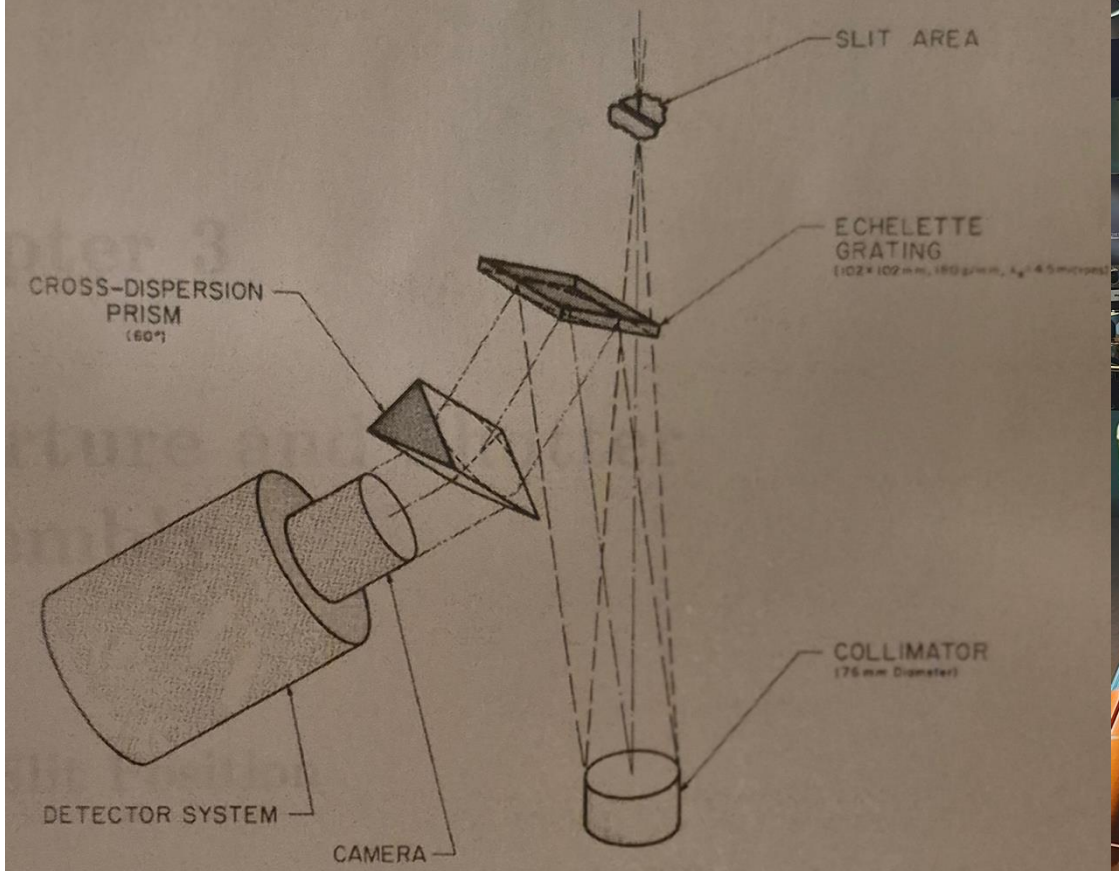
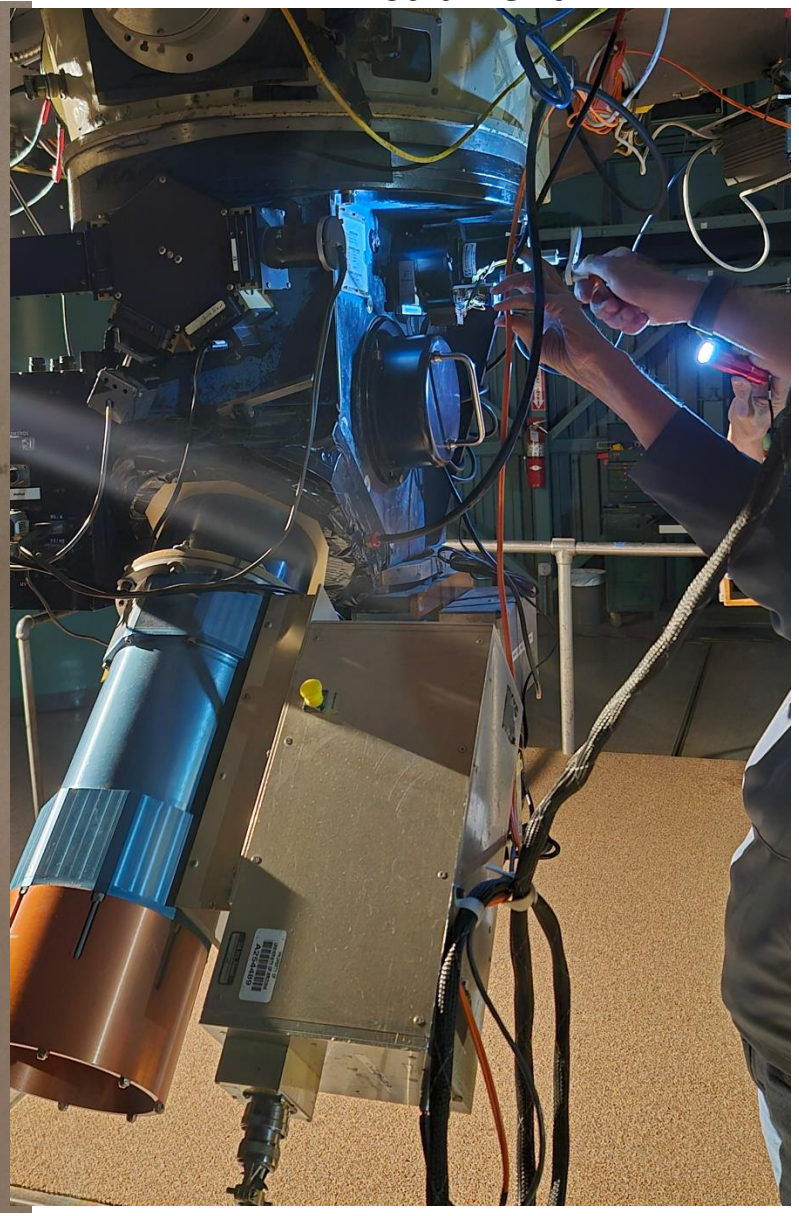


Figure 2.6: Schematic of echellette mode.

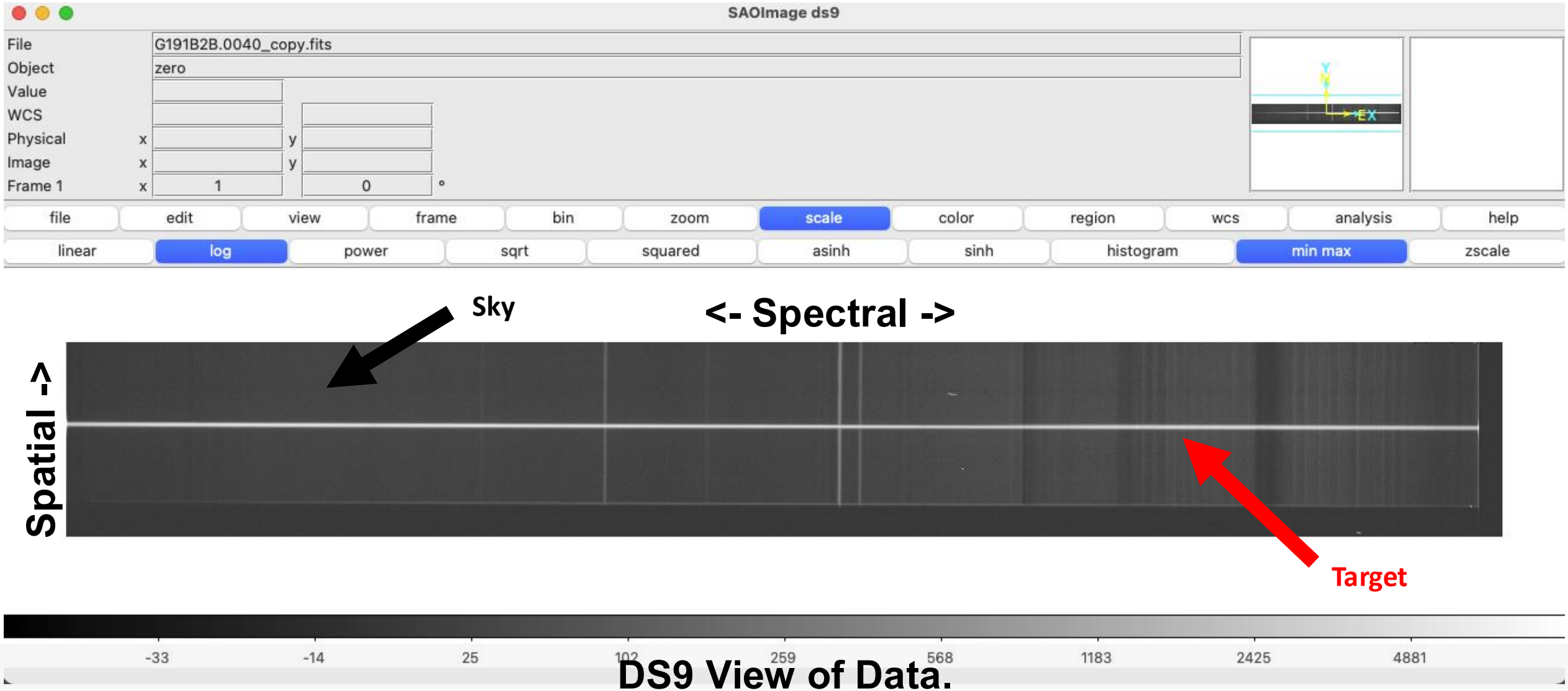
Instrument



Grating



Example Data: Boller & Chivens Spectrograph



DS9 View of Data.

Target: Standard White Dwarf

Reducing Spectroscopic Observations

Step 1: Create normalized flat field image (Fn).

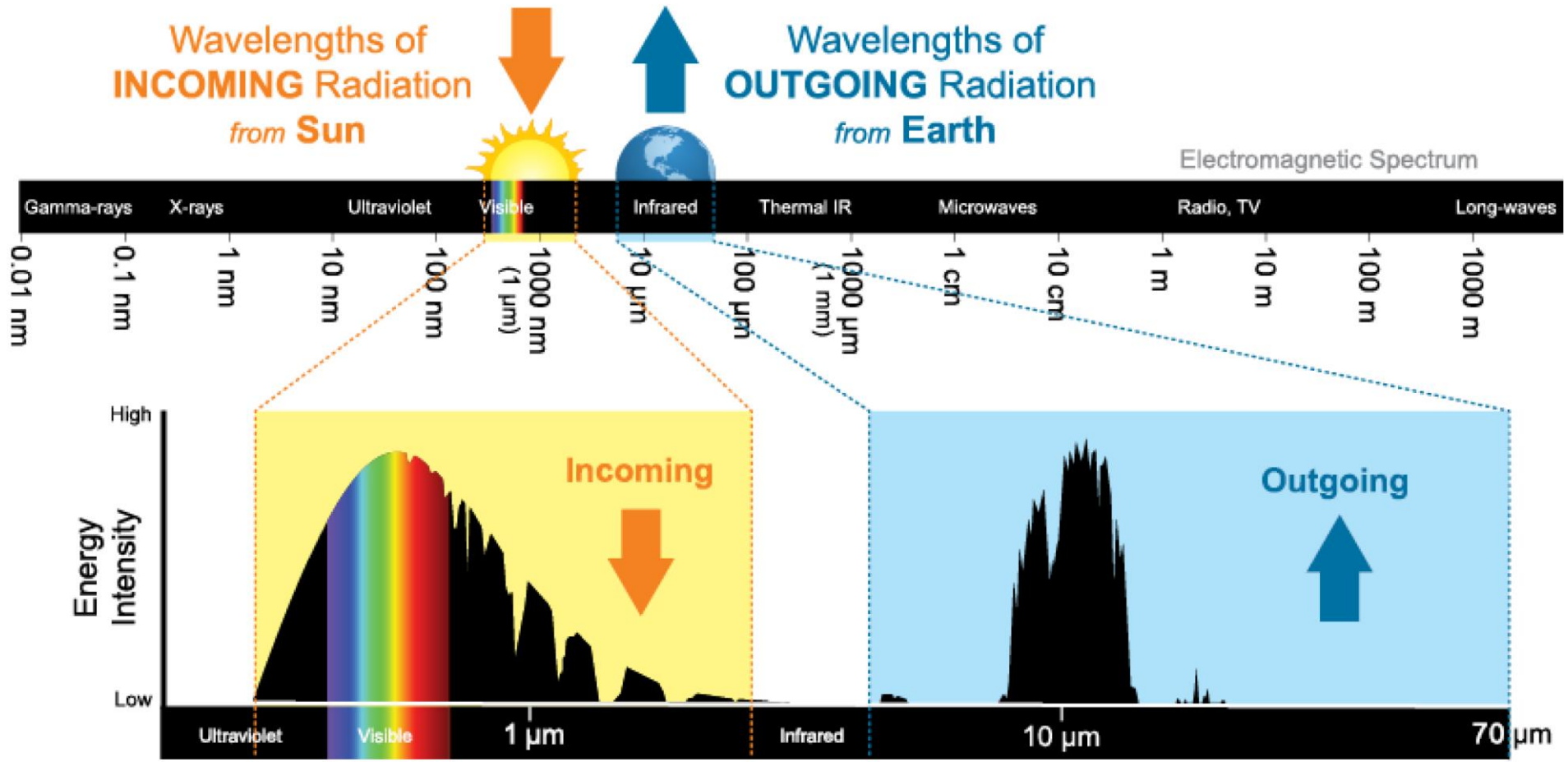
Step 2: (If necessary) scale dark image (D) to the same exposure time as your science image.

Step 3: Subtract the Bias image (B) and scaled dark image (sD) from your science exposure (S).

Step 4: Divide the adjusted science image by the normalized flat field to make a reduced scientific image.

Reducing spectroscopic data starts similar to imaging.

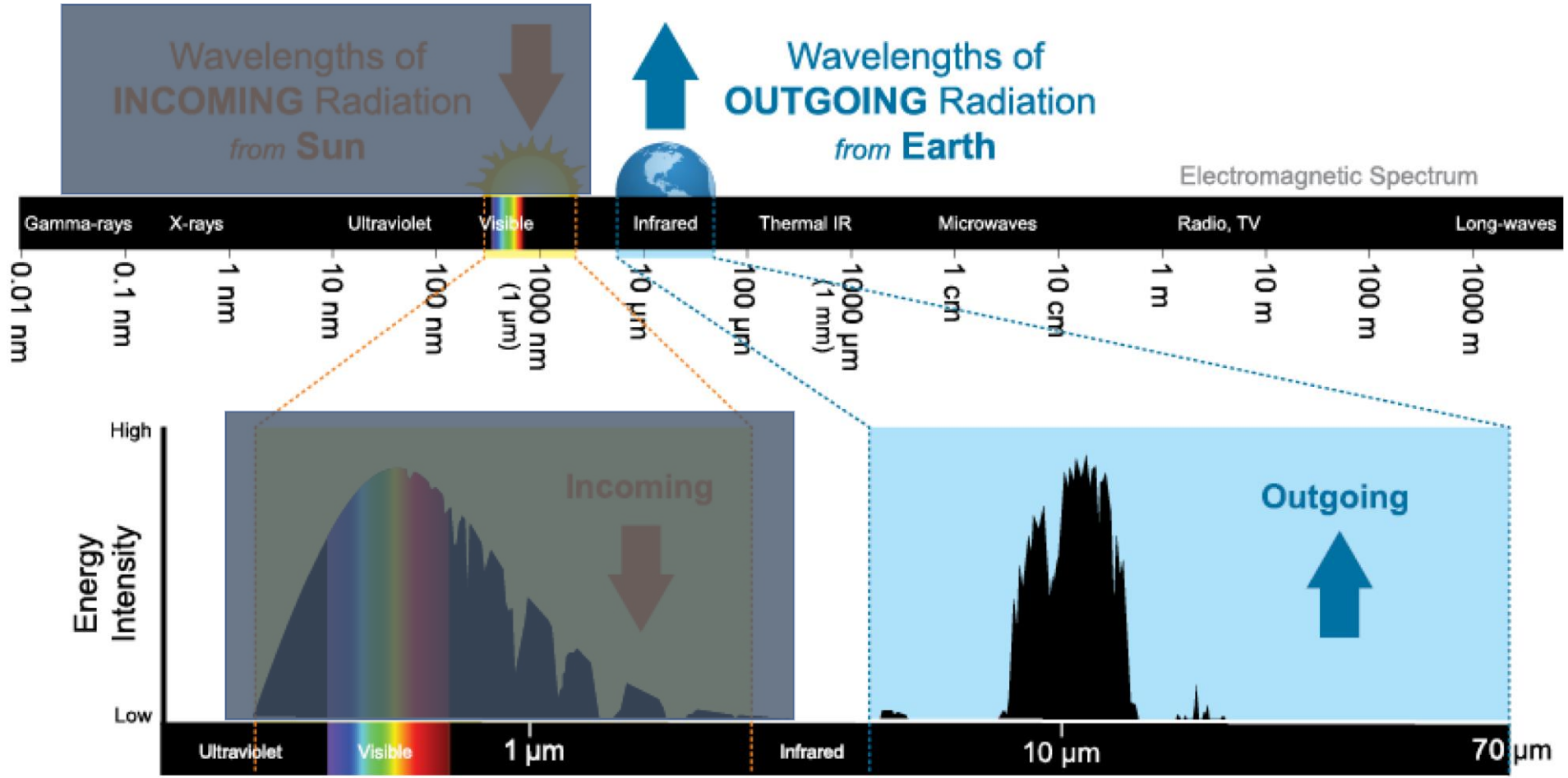
The Impact of Earth's Atmosphere: Sky Emission



Energy budget of Earth's atmosphere during the day time.

NOAA (day time)

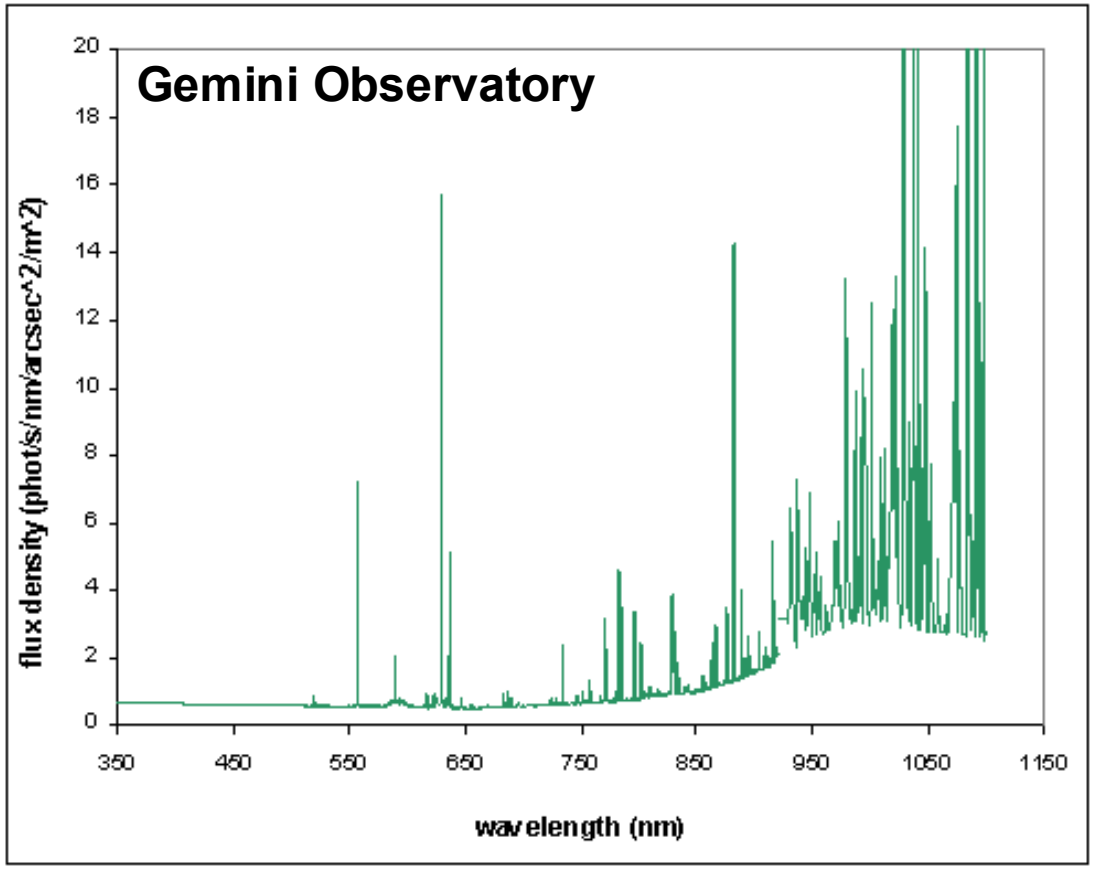
The Impact of Earth's Atmosphere: Sky Emission



Energy budget of Earth's atmosphere during night time.

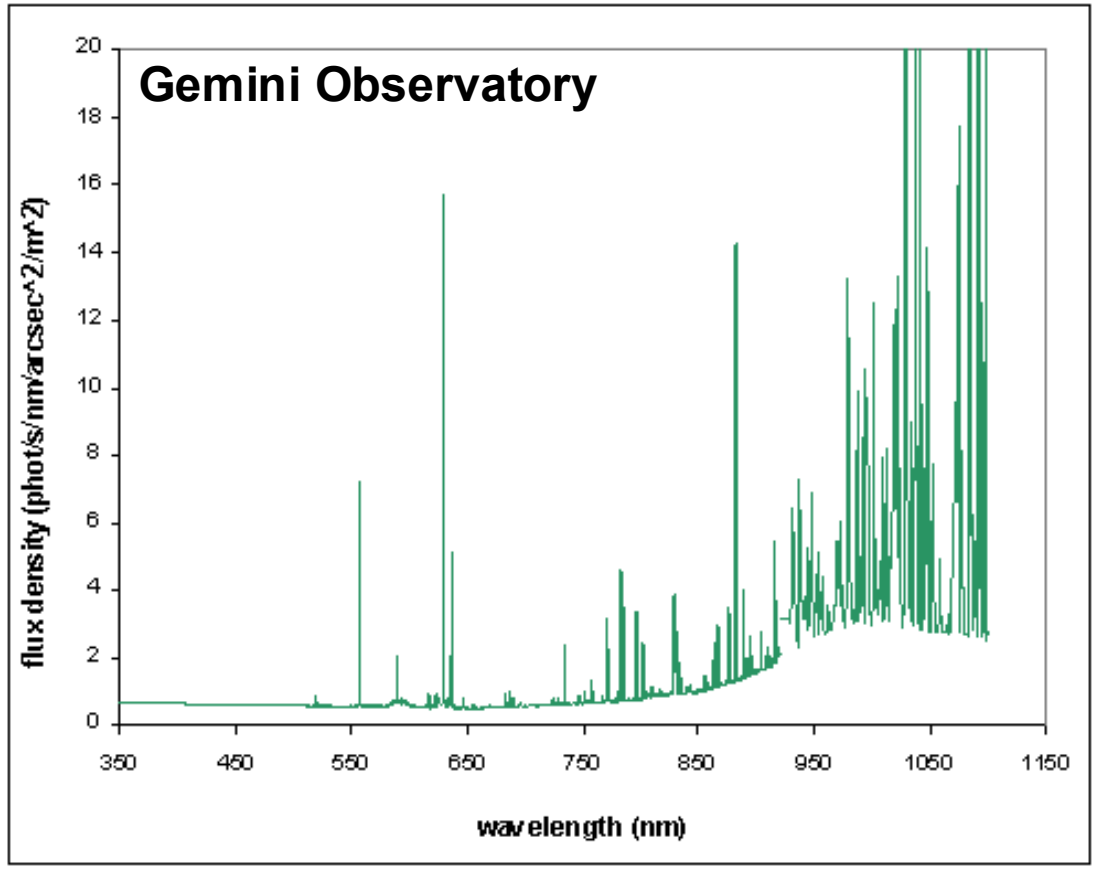
NOAA

Sky Background: Optical Wavelengths



- Earth's atmosphere has its own fingerprint that shows up in your data. This is impact is not removed by flat fielding, bias subtraction, or dark subtraction.

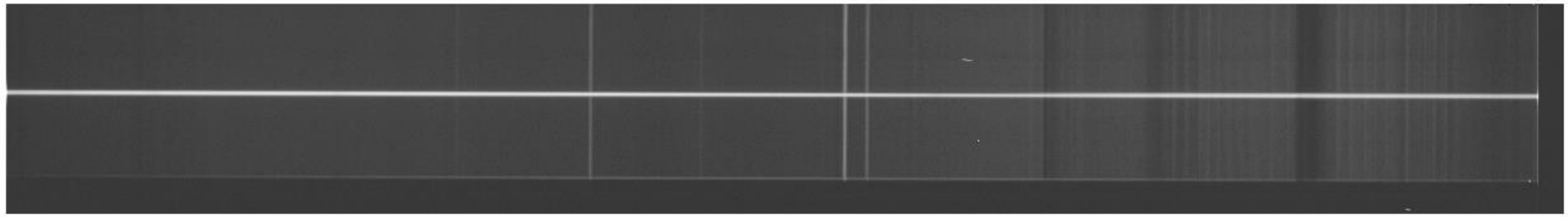
Sky Background: Optical Wavelengths



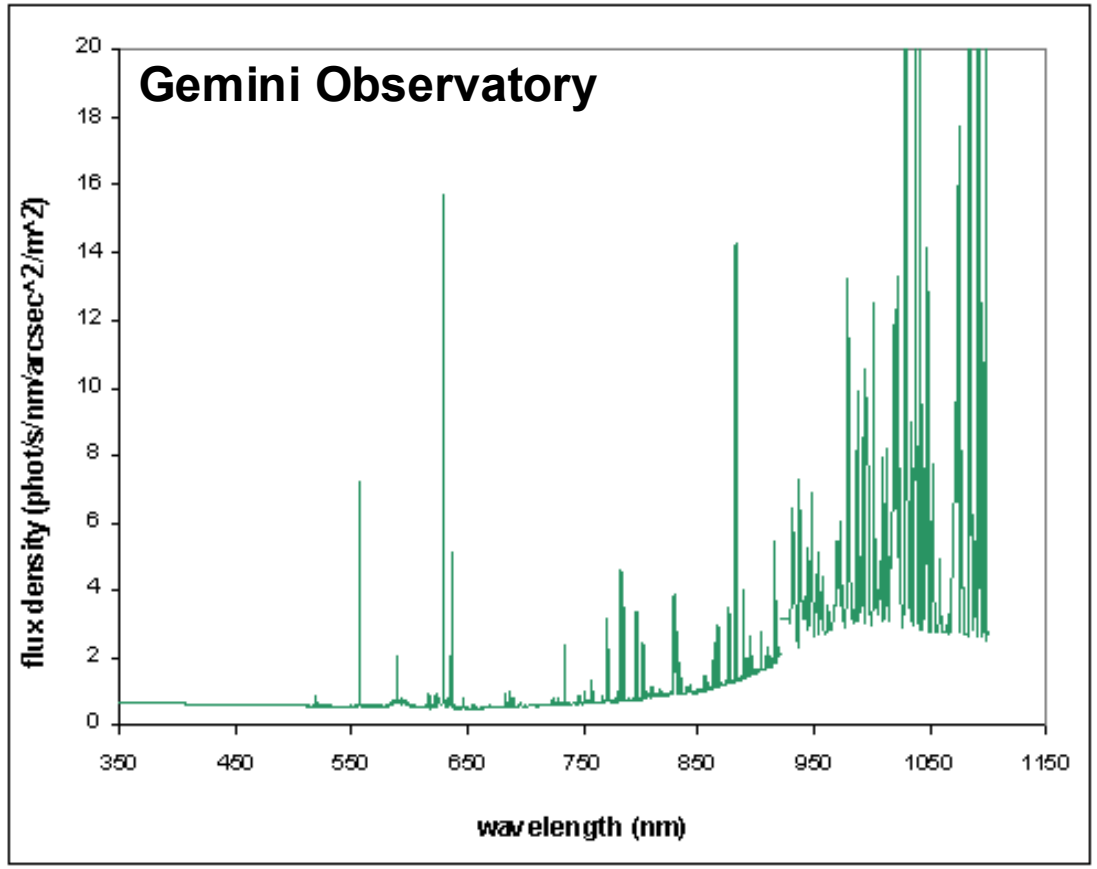
- Earth's atmosphere has its own fingerprint that shows up in your data. This is impact is not removed by flat fielding, bias subtraction, or dark subtraction.
- There are several ways to remove it:
  - Taking an average/median away from the target.
  - "Nodding" along the slit (IR)
  - Taking spectra of blank sky

<- Spectral ->

Spatial ->



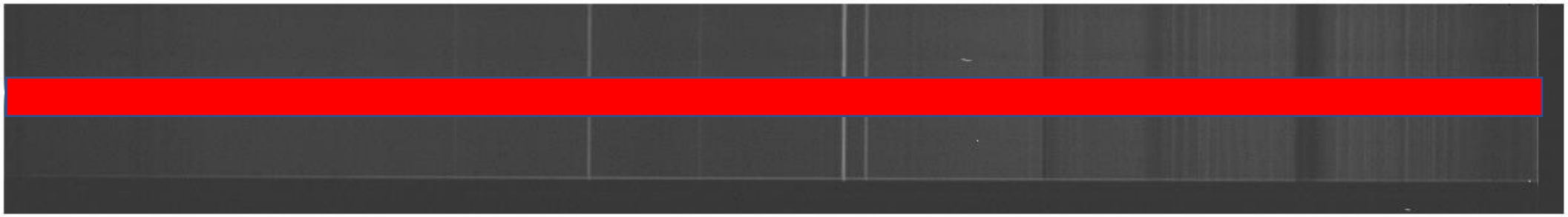
Sky Background: Optical Wavelengths



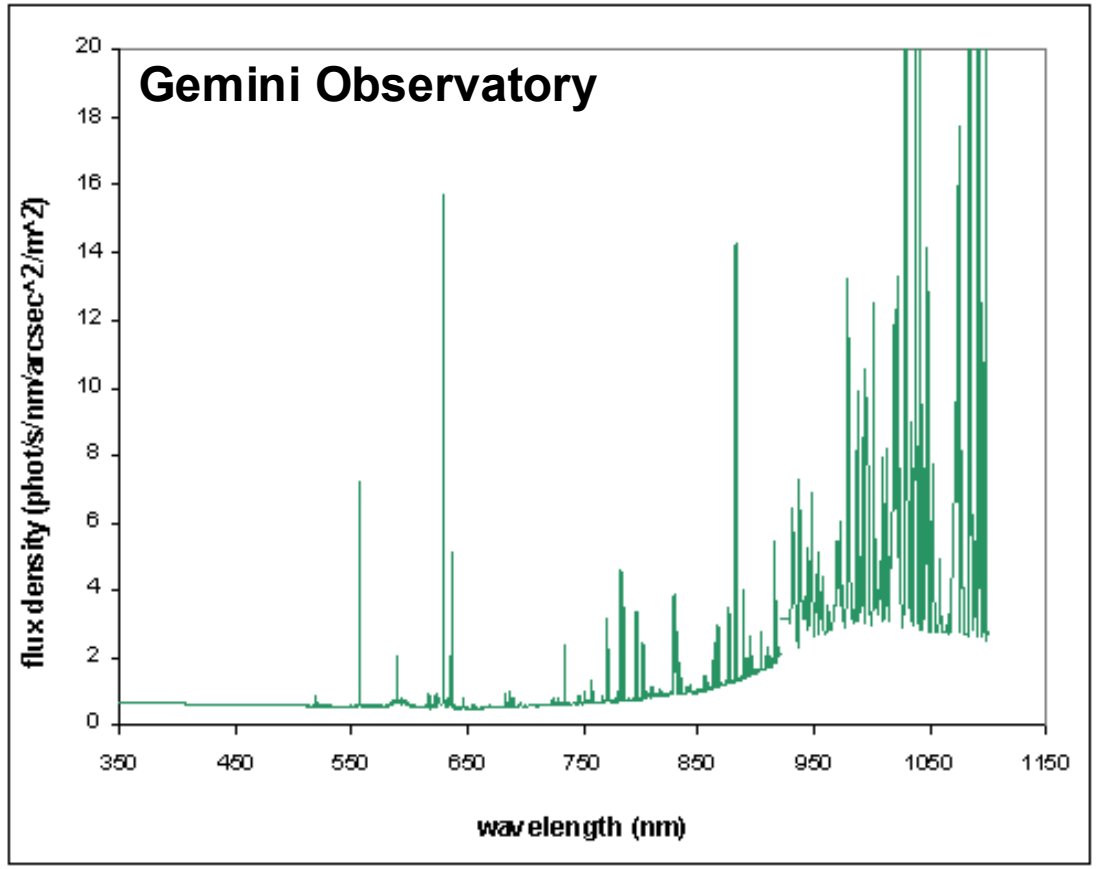
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Spatial ->



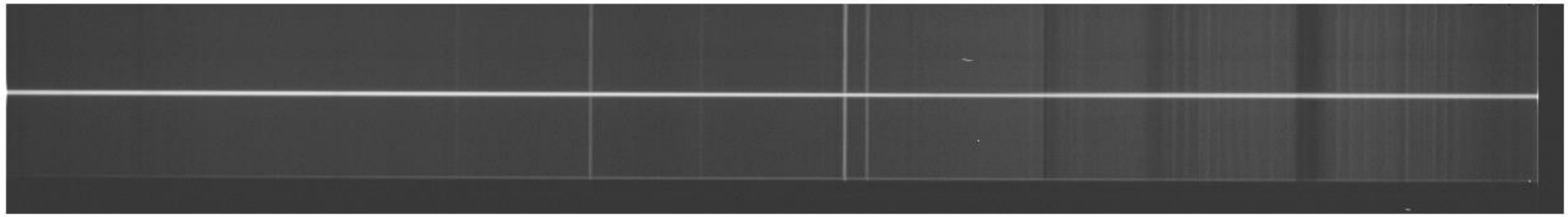
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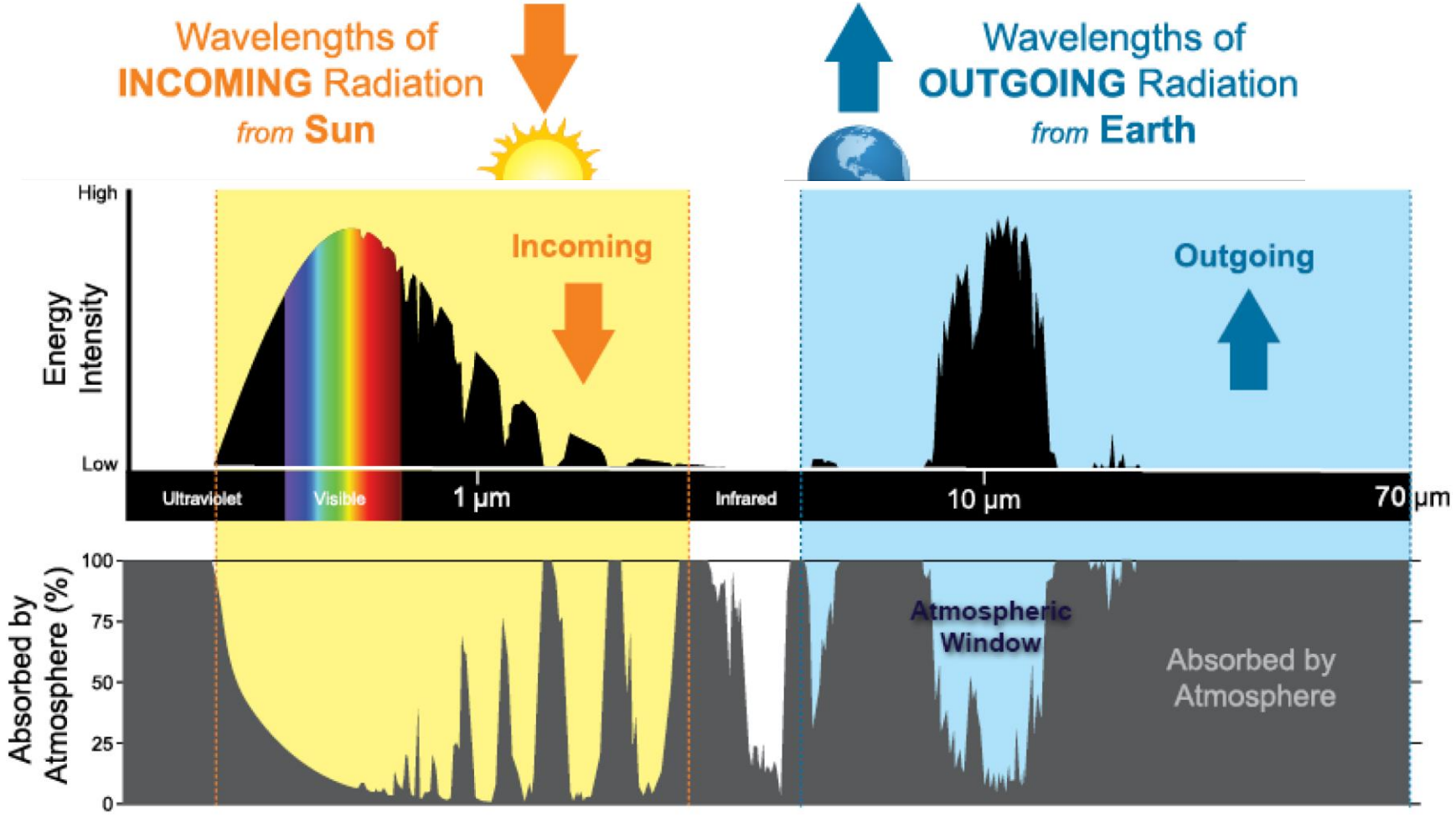
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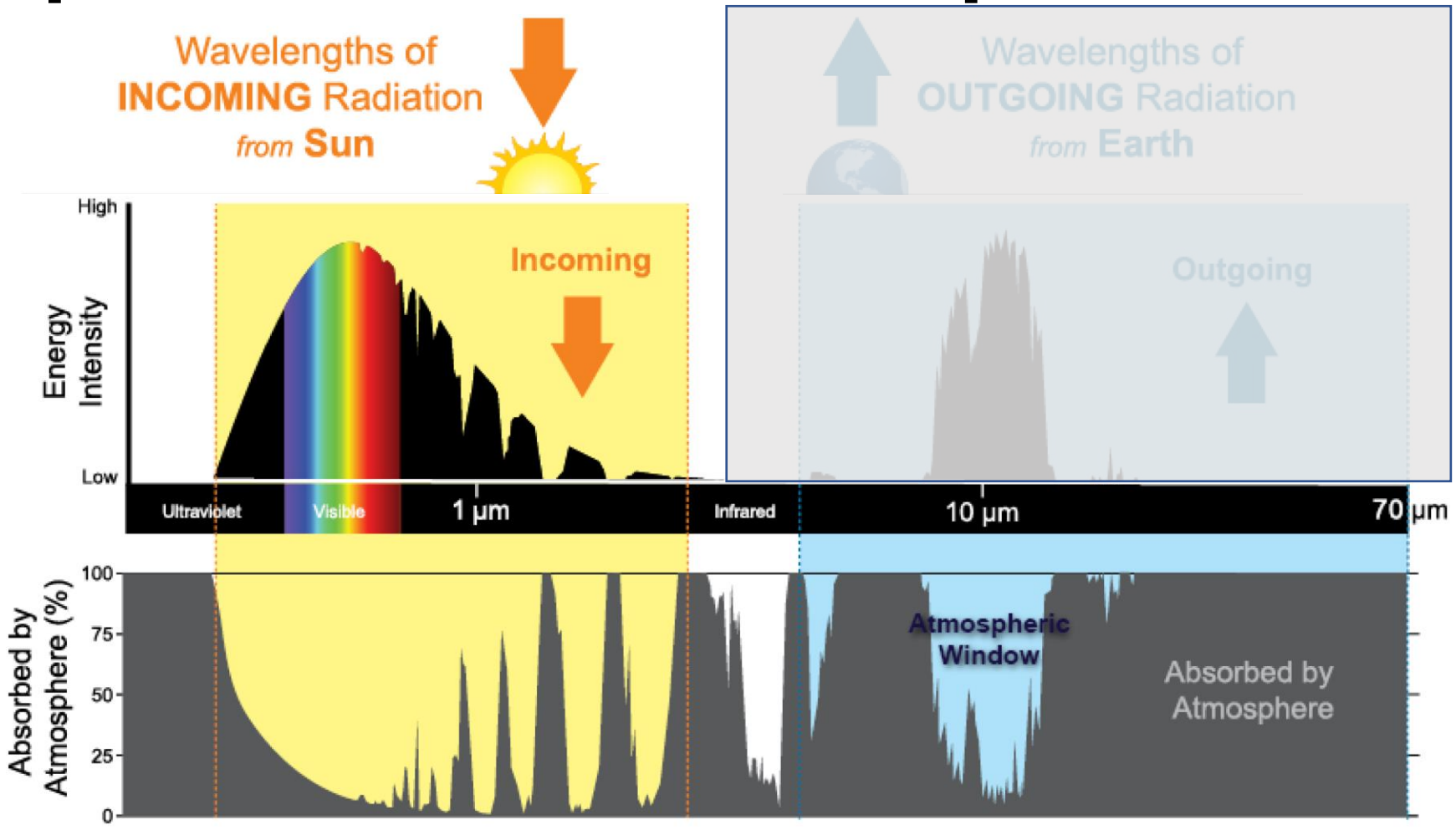


The Impact of Earth's Atmosphere: Absorption



Some of the absorption features apparent in your spectrum are not from your target, its caused by molecules in Earth's atmosphere. Removing this impact is often called **telluric calibration**.

The Impact of Earth's Atmosphere: Absorption



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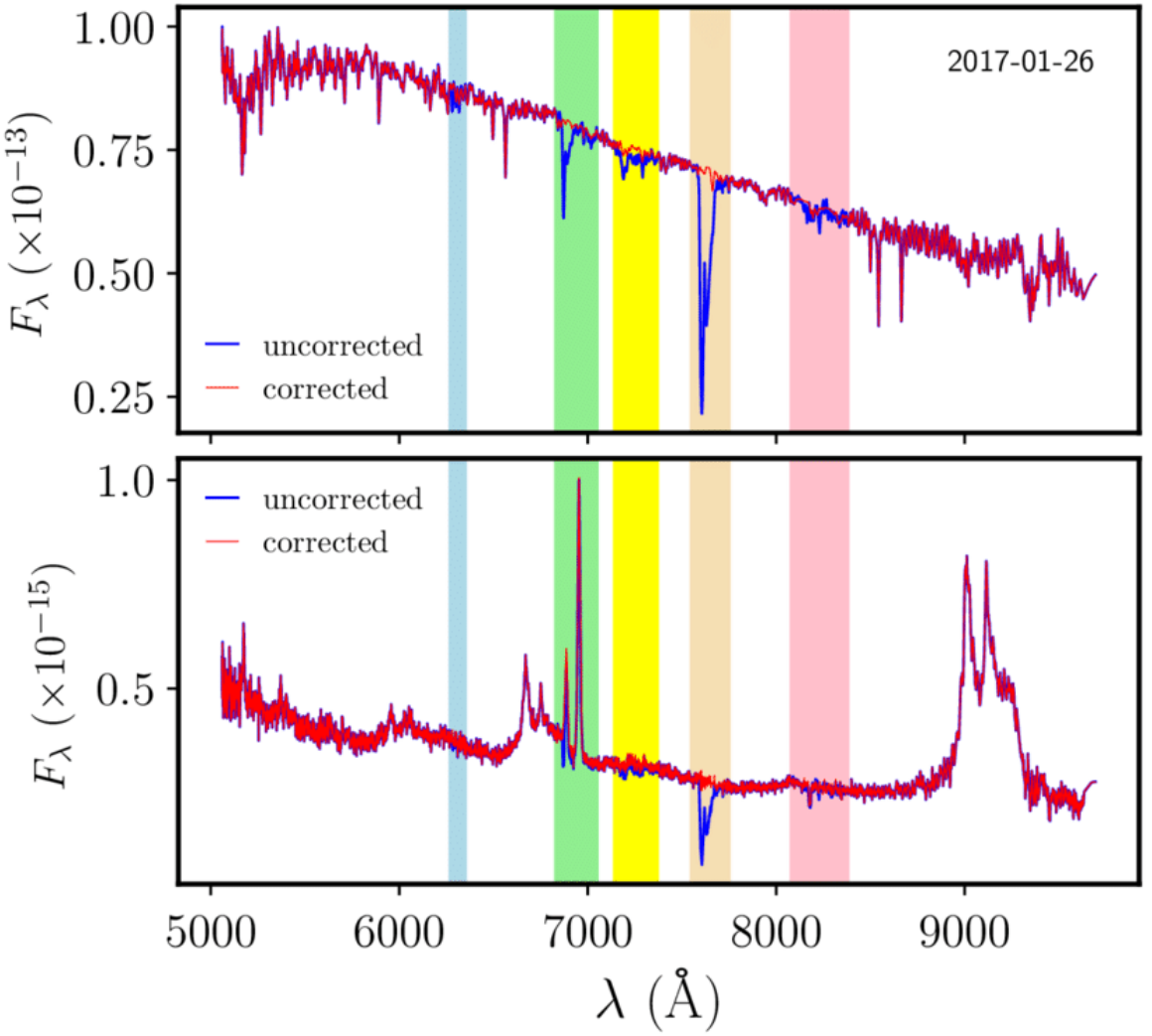
Telluric Calibrator Stars

- Telluric standard stars should have the following qualities:
  - Observed at similar airmass/atmospheric conditions as the science target(s).
  - Should not be variable.
  - Should not be secret binary star.
  - Ideally be brighter than your science observations.
  - They should have a “known” or simple spectral response like a black body ex: A0 stars, white dwarfs, some G type stars.
  - **Bonus:** It has a precise photometric point within the wavelength range of your data.

Telluric Calibration

Some of the absorption features apparent in your spectrum are not from your target, its caused by molecules in Earth's atmosphere. Removing this impact is often called **telluric calibration**.

$$F_{sci}(\lambda) = \frac{C_{sci}(\lambda)}{C_{cal}(\lambda)} * F_{cal}(\lambda)$$

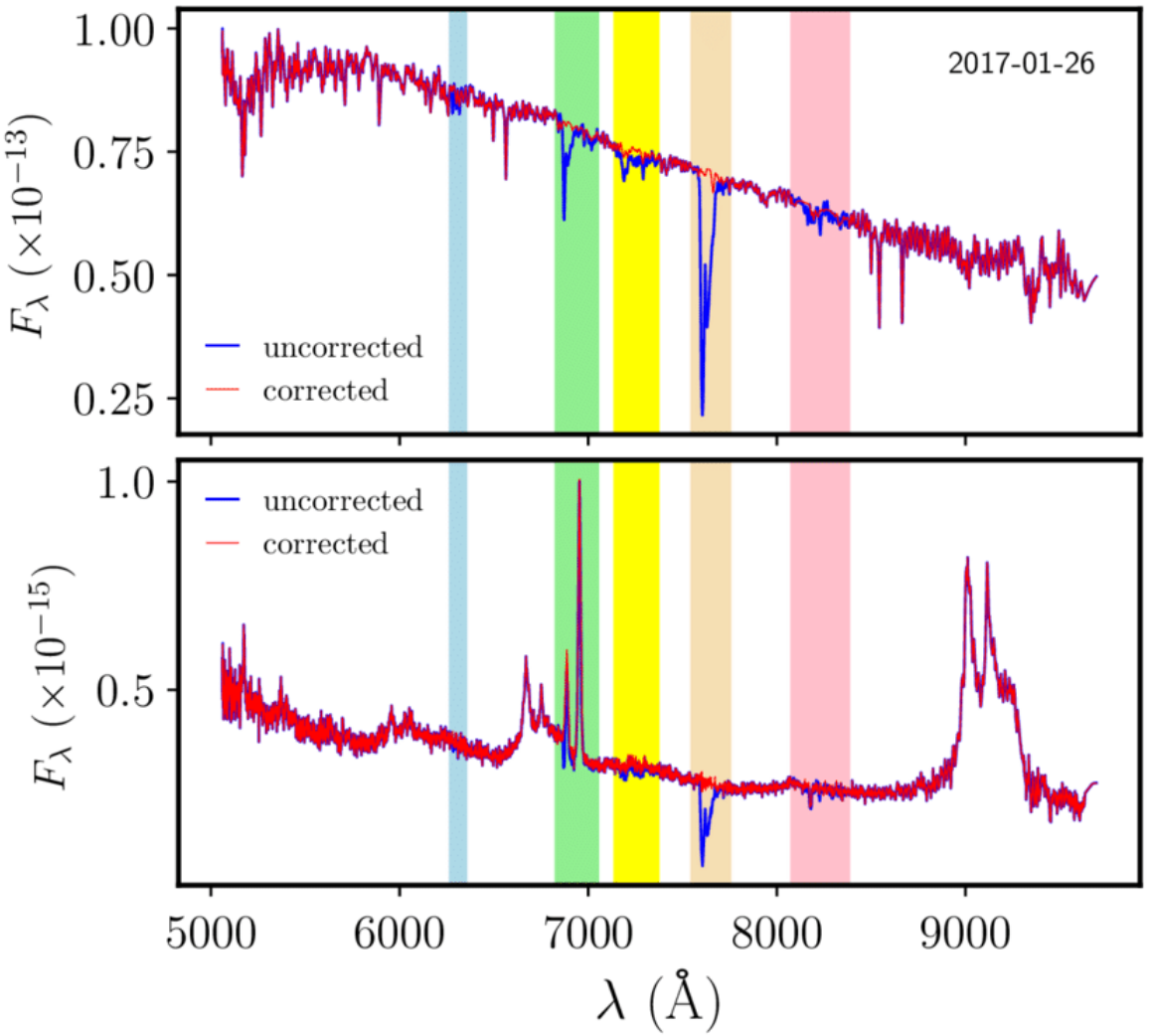


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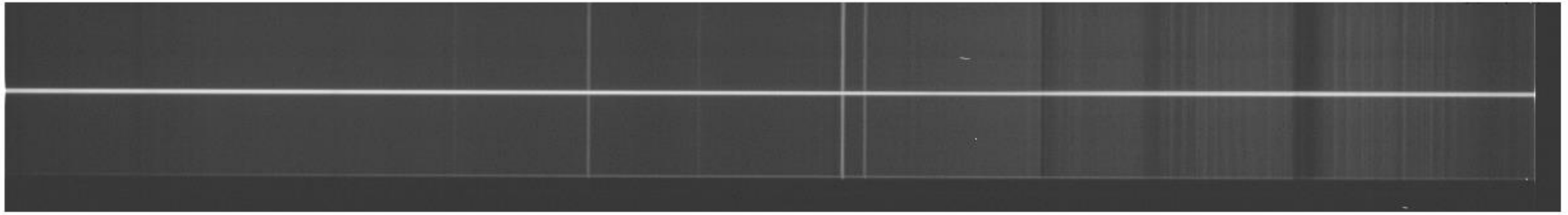
$$F_{sci}(\lambda) = \frac{C_{sci}(\lambda)}{C_{cal}(\lambda)} * B_{cal}(\lambda, T)$$



Wavelength Solution

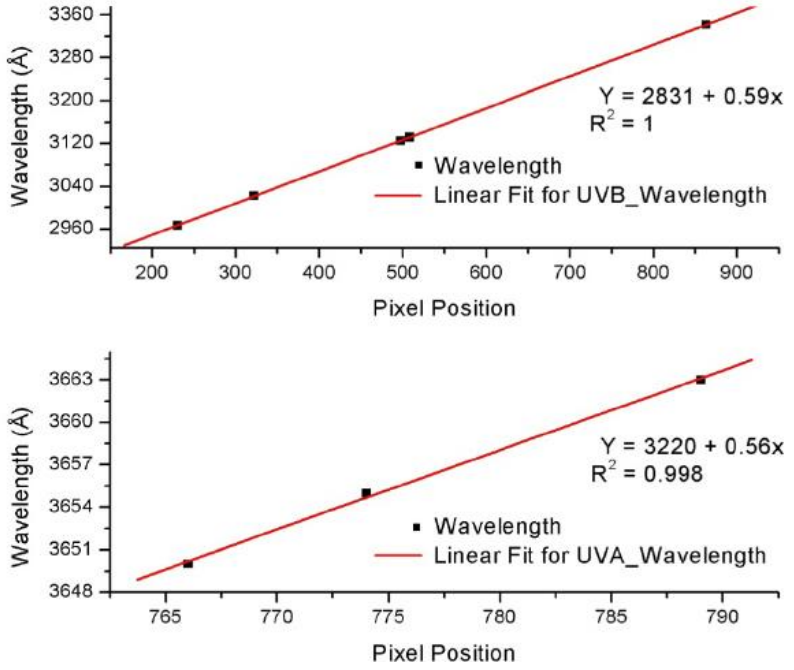
<- Spectral ->

Spatial ->



When you get your data, you do not automatically know what pixels correspond to each wavelength so you have to fit using your data or calibrations.

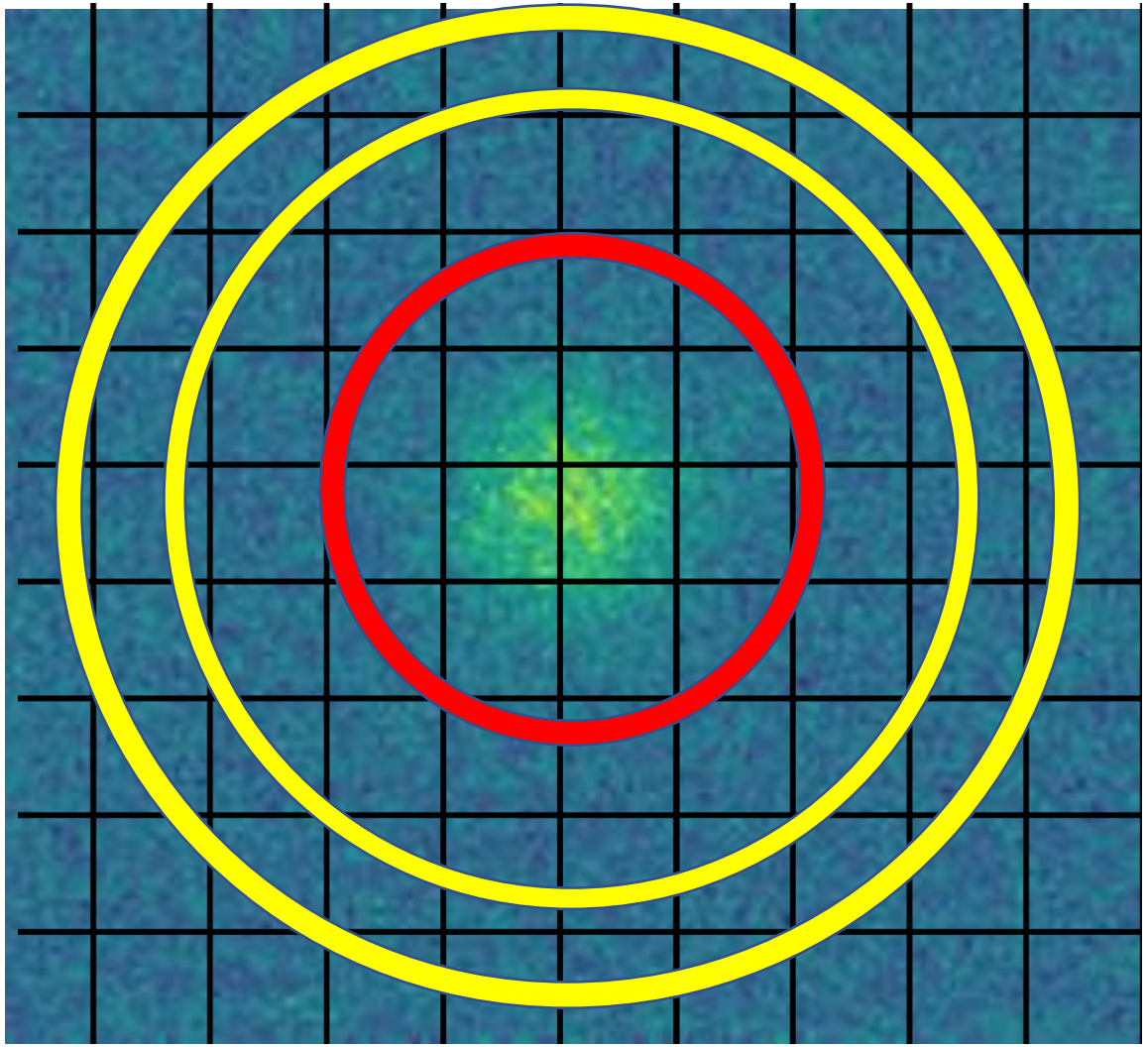
- Sky
- Arc lamps
- Laser Frequency Combs
- Many more options.

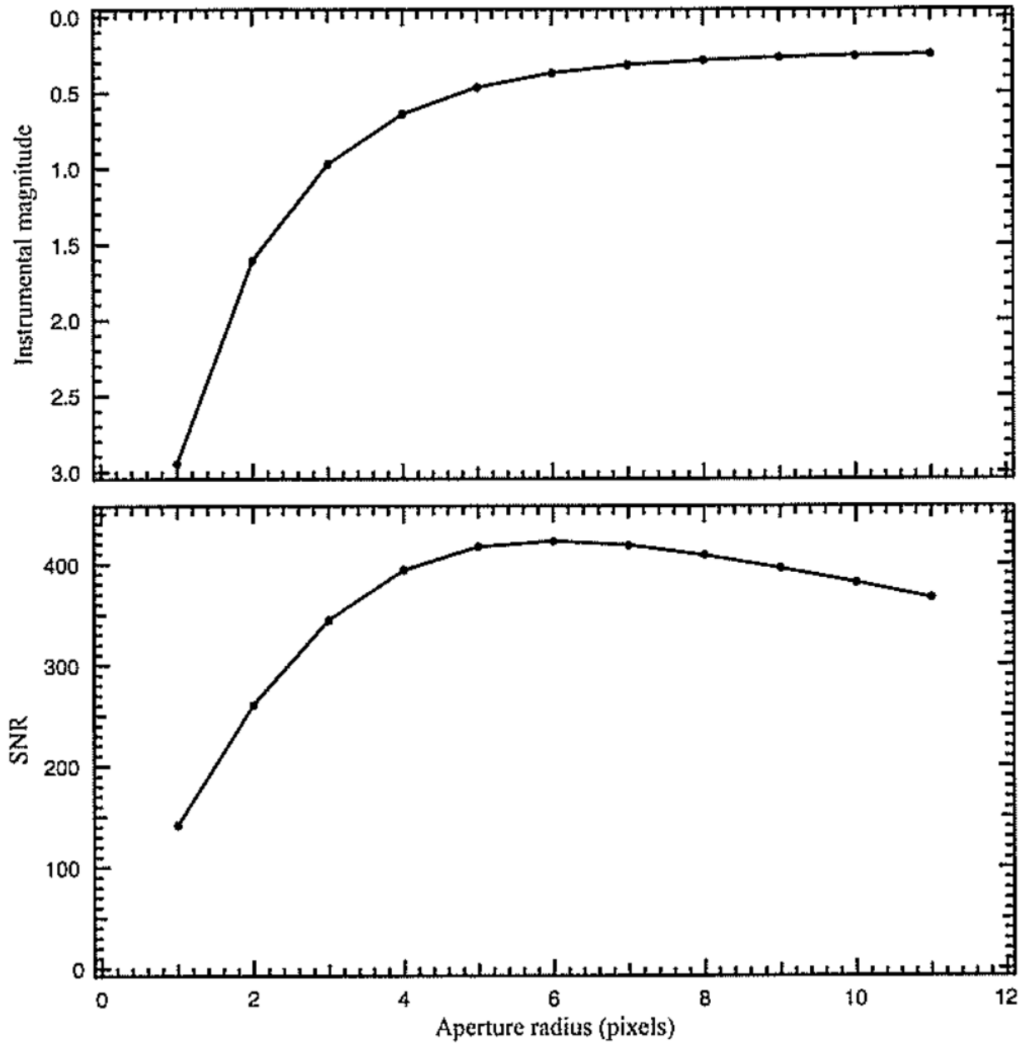


Example from google not B&C spectrograph

Aperture Photometry: What Radii Do I Pick?

Reduced Image



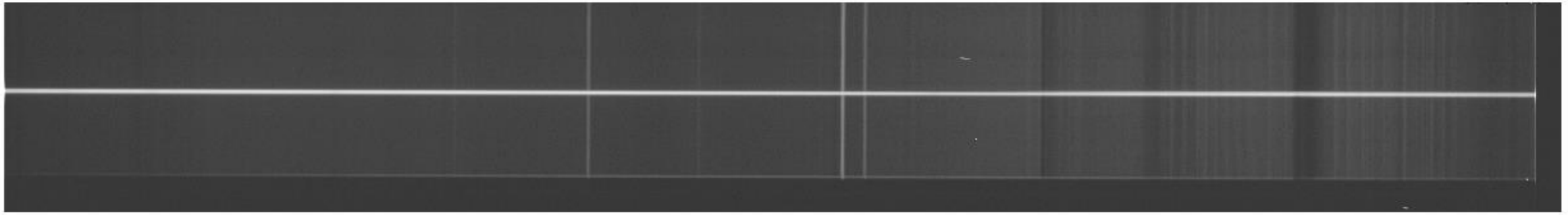


You usually choose an extraction radius to maximize signal to noise.

Aperture Extraction: What Radii Do I Pick?

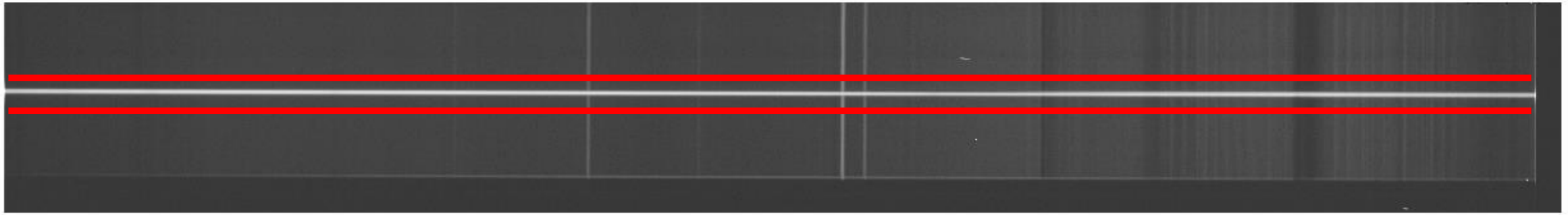
<- Spectral ->

Spatial ->



<- Spectral ->

Spatial ->

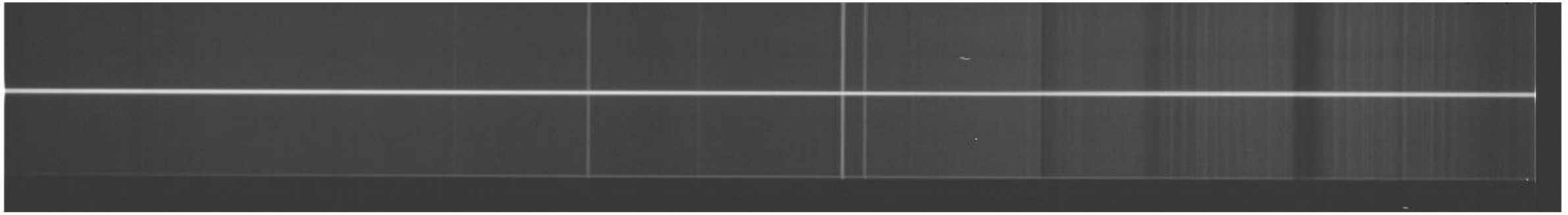


You add up light in the spatial direction to “extract” spectra.

Aperture Extraction: What Radii Do I Pick?

<- Spectral ->

Spatial ->



<- Spectral ->

Spatial ->



You add up light in the spatial direction to “extract” spectra.

You can take the standard deviation in the “background” (yellow) to estimate noise per wavelength